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# Flight delay propagation modeling: Data, Methods, and Future opportunities

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ABSTRACT

Persistent flight delays have continually posed a substantial challenge within the aviation sector. These delays infiltrate the air transportation network, precipitating disruptions to schedules and negatively impacting operational efficiency, passenger satisfaction, environmental considerations, and economic performance. Over the years, numerous methods have been developed for modeling flight delay propagation; however, a systematic review that encapsulates and synthesizes these approaches remains absent. This paper endeavors to examine the literature pertaining to flight delay propagation modeling, culminating in salient observations for future research potential. Initially, a comprehensive taxonomy framework is presented for delay propagation modeling. An extensive elucidation of data sources and application scenarios for various data types, two primary modeling perspectives, and an in-depth evaluation of seven modeling methods follows, accompanied by their characteristics and suitable application scopes. Subsequently, current research trends are summarized, and future research opportunities for modeling methods and research questions are detailed, establishing a solid foundation for continued exploration. The conclusions derived from these findings are anticipated to equip researchers, practitioners, and policymakers in the aviation industry with augmented comprehension of flight delay propagation patterns, thereby empowering them to prioritize resource allocation while enhancing overall system performance.

## 1. Introduction

The rise in air traffic demand and the expansion of highly connected air transportation networks have led to increased congestion and flight delays. Over the past decades, flight delays have become a significant concern in airport management and flight scheduling, impacting the efficiency of air transportation system operations and influencing passenger choices. In the US, it is estimated that the annual total delay cost surpasses $30 billion (Churchill et al., 2010). In 2018, nearly 2 million delayed flights in the US caused considerable disruptions to the normal operation of the air traffic system (Wang et al., 2022b). Numerous factors can precipitate flight delays and cancellations, including adverse weather conditions, air traffic control issues, airport or airspace closures, and flight operation challenges. Among these, the late arrival of preceding flights, known as delay propagation, is identified as the most critical cause of flight delays (Gopalakrishnan and Balakrishnan, 2021; Wang et al., 2022b).

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Flight delay propagation describes the phenomenon in which an initial delay experienced by a single aircraft can disseminate throughout the air transportation network. This often occurs because airlines typically optimize passenger movements when creating flight schedules, leading an aircraft to perform multiple flight legs according to a prearranged itinerary. Consequently, upstream flight delays may transmit to downstream flights, and due to the inherent connectivity of the network, this cascading delay effect permeates the entire aviation network.

The examination of flight delay and the delving into delay propagation mechanisms have long captivated scholars within the aviation domain, as well as operational managers in the industry (Rebollo and Balakrishnan, 2014; Kafle and Zou, 2016; Guo et al., 2022). Efficacious and highly precise flight delay predictions facilitate the anticipation of flight operation dynamics, empowering aviation authorities to mitigate delays and curtail losses attributable to such disruptions (Yu et al., 2019). In comparison to delay prediction research, delay propagation inquiries may prove more enticing and challenging, as they enable the identification of delay origins, the computation of reactionary delays, and the comprehension of delay evolution. Such understanding paves the way for implementing potent measures to counteract the cascading effect of delays (Cai et al., 2021b).

Throughout the past few decades, researchers have devised a range of approaches to model flight delay propagation, with the goals of understanding the underlying mechanisms, identifying contributing factors, and formulating effective delay mitigation strategies. These models encompass statistical and analytical methods (Wong and Tsai, 2012; Kafle and Zou, 2016), as well as advanced machine learning and data-driven techniques (Bao et al., 2021; Cai et al., 2021a; Lin et al., 2021), each possessing its own strengths and weaknesses. Furthermore, the developmental trajectory in delay propagation modeling has transitioned from deterministic modeling approaches to those that incorporate uncertainty and dynamics (Wu and Law, 2019; Li and Jing, 2021). Simultaneously, research has progressed from examining delays at individual airports or flights to modeling delay propagation within air transportation networks (Zanin et al., 2017; Du et al., 2018; Cai et al., 2021b; Guo et al., 2022).

Despite extensive efforts on flight delay propagation mechanisms conducted by scholars across diverse fields, including aviation, transportation, computer science, management science, statistical science, and complex systems, inconsistencies may remain in problem descriptions, underlying assumptions, and scenario definitions across these domains. Currently, there is an absence of a comprehensive review that systematically consolidates the various research models and methods, comparing their characteristics and applicability. Moreover, a unifying framework encompassing disparate data types, their applications, and perspectives on modeling flight delay propagation, along with an overview of current research trends and potential future research opportunities, is yet to be established. These existing gaps may impede the advancement of delay propagation modeling in both theoretical and practical contexts.

This paper endeavors to present a comprehensive review of flight delay propagation modeling, encompassing data applications, modeling perspectives, modeling methods, research trends, and potential future research opportunities. To the best of our knowledge, this constitutes the first attempt to offer a holistic review of flight delay propagation modeling. The primary objective of this review is to introduce data and modeling approaches, thereby underlining potential promising research opportunities for future endeavors. The specific research questions we seek to address through this study are as follows:

(1) Which data types can be utilized for investigating flight delay propagation, their application scopes, and their usage frequency in different air transportation systems?
(2) What methods are available for modeling flight delay propagation within a certain problem perspective, and what are their characteristics and applications?
(3) What are the current research trends and prospective future research opportunities in modeling flight delay propagation?

We should note that there have been three recent surveys related to flight delay, each with distinct focus areas and minimal overlap with our work. Zhixing et al. (2021) examine flight delay from a complex network perspective, describing the relationships between static air traffic networks, their temporal evolution, and associated network metrics. However, it addresses flight delay propagation only in terms of cascade failures or epidemic processes within air traffic networks. Wang et al. (2022a) review research on flight delay prediction, analyzing methods ranging from traditional statistical analysis and simulation modeling to machine learning. Yet, it does not extensively discuss issues related to flight delay propagation. Carvalho et al. (2021) explore flight delay research from a data science viewpoint, discussing various types of delays, data sources, processing methods, and data analysis techniques within a scattered framework. However, it also lacks a detailed analysis and summary of specific data sources, applications, modeling methods, research trends, and future opportunities in flight delay propagation problems.

Compared with previous surveys on flight delay, our survey stands out with the following key contributions:

- **Focused and In-depth Approach to Delay Propagation:** Our manuscript uniquely addresses the nuances of delay propagation with a targeted and profound analysis, allowing for a deeper understanding of the subject.
- **Comprehensive Methodological Overview:** We bridge the gaps left by previous reviews by including a thorough exploration of methodologies not fully covered before, such as Bayesian networks, queueing theory, and economic analysis.

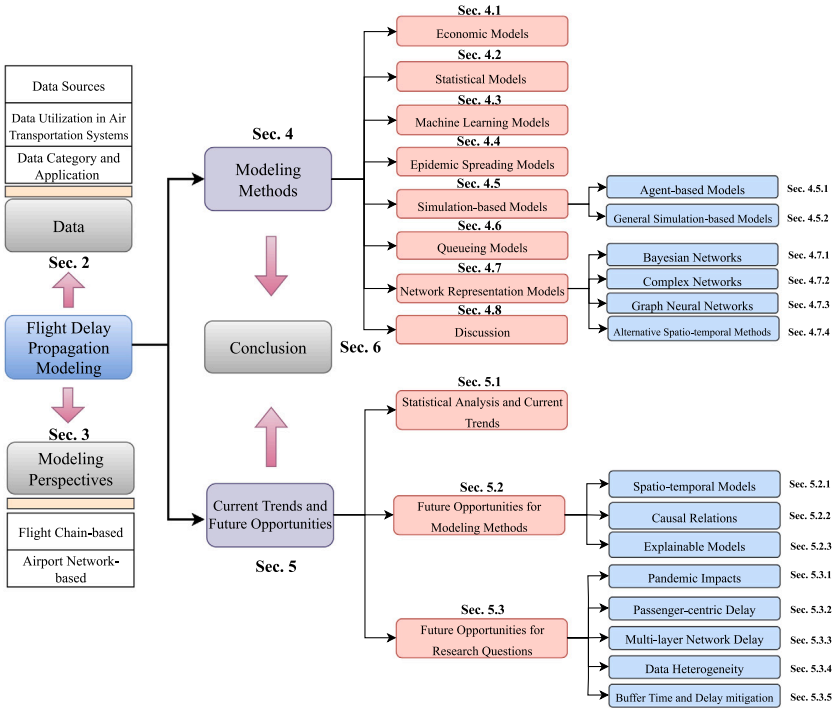



**Fig. 1.** Taxonomy framework of flight delay propagation modeling.

- **Detailed Data Source Analysis:** Our work advances beyond simply listing data sources by demonstrating their application in various air transportation systems. We delve into the scope of data application, exploring how different data types can be synergized with diverse research methodologies.
- **Extensive Literature Analysis with Comparative Tables:** Our review enhances the standard literature summary by conducting detailed analyses of research methods and presenting these in comparative tables. This approach offers a multi-dimensional perspective that enriches the existing body of knowledge.
- **Future Research Perspectives:** Distinguishing our survey from others, we provide a forward-looking perspective, highlighting specific insights and detailed literature references to explore the potential applications of advanced technologies and address unresolved issues in the field.

Besides this introduction, the remainder of this paper is structured as follows. Section 2 introduces the data sources and their utilization in different air transportation systems, along with their application scopes in flight delay propagation modeling. Section 3 delineates two main modeling perspectives. In Section 4, we explore various flight delay propagation modeling methods employed by researchers and provide a detailed discussion on the characteristics of different methods, influencing factors, primary challenges, and frontier technologies. Section 5 examines current trends and outlines future research opportunities. Finally, Section 6 concludes the paper, emphasizing the significance of our study for researchers and practitioners in flight delay propagation modeling. We present the overall taxonomy framework and the relationships between sections within this review in Fig. 1.

## 2. Data

In our review of the existing literature on flight delay propagation modeling, we conducted searches in Google Scholar and Scopus databases until September 2022 using keywords related to delay propagation modeling, such as “flight delay propagation modeling”, “flight delay propagation”, “propagation modeling”, and “delay propagation”. We initially identified relevant literature and then proceeded to filter the articles. As our research focuses on flight delay propagation modeling, we excluded articles that solely analyzed the impact and causes of flight delays or centered on flight delay prediction. We also filtered out articles with low citation counts or non-English literature. Furthermore, we performed a secondary screening by reading related literature and examining cited references and citing articles. Ultimately, we confirmed 98 articles for analyzing modeling methods and an additional 22 articles for exploring future research opportunities.

**Table 1**
Flight delay data sources by frequency of use.

| Data source | Type | Publicly available | Url | Frequency |
|---|---|---|---|---|
| Bureau of Transportation Statistics (BTS) | Flight operation data | ✓ | https://www.transtats.bts.gov | 25 |
| Aviation Operations Department | Flight operation data | NA | / | 23 |
| Aviation Authority | Flight operation data | NA | / | 17 |
| Aviation System Performance Metrics (ASPM) | Flight operation data | Partially | https://aspm.faa.gov/apm/sys/AnalysisAP.asp | 12 |
| Airline Service Quality Performance (ASQP) | Flight operation data | ✓ | https://aspm.faa.gov/asqp/sys/ | 7 |
| Eurocontrol R&D data archive | Flight operation data | Request | https://www.eurocontrol.int/dashboard/rnd-data-archive | 6 |
| VariFlight | Trajectory data | Partially | https://www.variflight.com/en | 5 |
| Enhanced Traffic Management System (ETMS) | Flight operation data | Request | https://www.fly.faa.gov/Products/Information/ETMS/etms.html | 4 |
| National Oceanic and Atmospheric Administration (NOAA) | Weather data | ✓ | https://www.ncdc.noaa.gov/cdo-web/datatools/lcd | 2 |
| Eurocontrol Central Office for Delays Analysis (CODA) | Flight operation data | Partially | https://ansperformance.eu/acronym/coda/ | 2 |
| Weather Underground | Weather data | Partially | https://www.wunderground.com | 2 |
| Flightradar24 | Trajectory data | Partially | https://www.flightradar24.com | 2 |
| ADS-B data | Trajectory data | NA | / | 2 |

The Aviation Operations Department refers to data originating from airports or airlines in specific countries and regions.
The Aviation Authority refers to data sources from specific national aviation management departments, such as the Civil Aviation Administration of China (CAAC), the National Civil Aviation Agency (ANAC) in Brazil, Research and Innovative Technology Administration (RITA), Aviation Data Communication Corporation (ADCC), Civil Aviation Authority of Singapore (CAAS), and so on.
ADS-B data refers to the data mentioned in the literature where the specific source is not stated. Therefore, here we cannot provide detailed websites and related information for such data.
"✓" means that the data is completely open to the public and can be directly downloaded from the corresponding website; "Partially" denotes that part of the data is open to the public, but complete historical information requires additional application; "Request" means that the data is only available to website members or specific institutions or users; "NA" indicates that the data information is not public.

### 2.1. Data sources

Table 1 displays the data sources used more than once in studies related to modeling flight delay propagation, along with their access links and frequency of use.[1] Notably, the Bureau of Transportation Statistics (BTS) data emerges as the predominant source, succeeded by data from the Aviation Operations Department and Aviation Authority. It is crucial to clarify that the "Aviation Operations Department" and "Aviation Authority" do not refer to singular data sources; typically, these encompass proprietary data not publicly accessible. Our data compilation often includes specific airlines and airports, with some studies withholding precise sources due to data privacy. To facilitate the presentation of data usage frequency, we have categorized such data under the "aviation operations department". Similarly, "Aviation Authority" denotes official national entities like China's Civil Aviation Administration (CAAC) and Brazil's National Civil Aviation Agency (ANAC). The primary preference for BTS data stems from its unrestricted public accessibility, without the need for additional applications or fees. Furthermore, apart from flight operation data, weather data and flight trajectory data are also frequently used in this field of research. The National Oceanic and Atmospheric Administration (NOAA) data, Weather Underground data, and Variflight data are particularly notable in these categories.

Table 1 additionally presents information regarding the accessibility of these data sources. We have delineated four categories of accessibility, which can be found in the footnote beneath Table 1. It is evident that the Airline Service Quality Performance (ASQP) data is readily available, being directly accessible from the BTS database website. For certain datasets, such as Aviation System Performance Metrics (ASPM) and Variflight data, limited data is openly accessible, while access to comprehensive historical

[1] In our analysis, articles often used multiple data sources and covered various air transportation systems, with diverse descriptions of flights and airports and multiple levels of delay propagation characterization. Hence, data usage frequencies are cumulatively tallied across categories, resulting in a total frequency that may exceed the number of articles reviewed. This cumulative counting principle is consistently applied in the frequency analyses of subsequent chapters, and thus will not be reiterated in later sections.

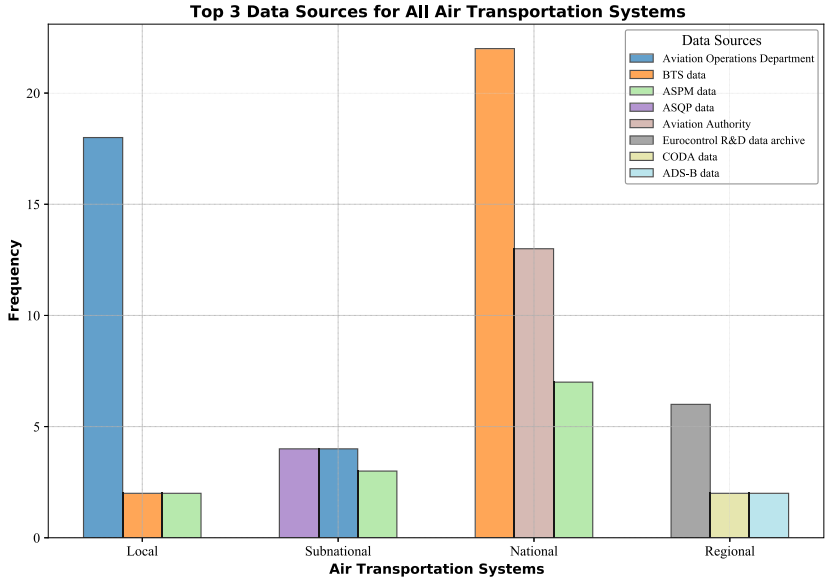



**Fig. 2.** Comparison of top three data sources in different air transportation systems.

datasets or detailed field data requires additional applications. Additionally, a range of weather data, including those from the National Oceanic and Atmospheric Administration (NOAA), can be obtained with ease. However, access to specific datasets like the Enhanced Traffic Management System (ETMS) and the Eurocontrol R&D data archive necessitates specific user credentials or further applications. With these varied data sources and their associated information, researchers are equipped to select the most suitable data, or a combination thereof, in alignment with their research goals and the specific context of their studies.

### *2.2. Data utilization in air transportation systems*

In this section, we delve into the modeling of flight delay propagation within different air transportation systems, offering diverse perspectives on this phenomenon. Our discussion encompasses an analysis of the frequency of data source usage across various systems, coupled with an examination of the national composition within these systems. Additionally, we provide statistical analyses and distributional insights into flights, airports, and airlines specific to each system. This comprehensive approach not only facilitates a granular dissection of data utilization in different air transportation systems but also aims to offer valuable insights. These insights could potentially aid researchers and practitioners in their endeavors within the field, enhancing understanding and application of data in the context of flight delay propagation modeling.

In our study, air transportation systems are categorized into four distinct types based on the geographical scope of their research: Local, Subnational, National, and Regional. The "Local" system pertains to studies focusing on individual airports or specific airlines, offering micro-level insights into their operations. "Subnational" systems refer to research confined to a specific region within a country, such as the Greater Bay Area. This encompasses multiple airports and operators but does not cover the country's entire air traffic system. The "National" category is for studies addressing the air transportation system of an entire country, providing a comprehensive view of its domestic air traffic. Finally, "Regional" systems are defined as those studies that transcend national boundaries yet are confined to a specific geographical region, such as the European and Southeast Asian airspace.

Fig. 2 provides an overview of the prevalent data sources within four air transportation system categories, revealing distinct preferences in data utilization. "Aviation Operations Department" data is most frequently cited in local systems, indicating a significant reliance at this granular level. Conversely, "BTS data" is favored in national systems, suggesting a strategic preference for broader transportation statistics. Across all categories, "ASPM data" emerges as a common source, reflecting its widespread acceptance for air traffic analysis. This visualization underscores the diversity of data source integration across the spectrum of air traffic systems, with certain sources being pivotal at multiple tiers of air traffic management.

Fig. 3 elucidates the distribution of country-specific research across different air transportation systems. The chart reveals that the national system exhibits the highest research frequency, with a notable predominance of studies from the USA and China. Conversely, the regional system shows a more limited range of countries, with Europe and Southeast Asia taking the lead. In studies focusing on the propagation of flight delays within local systems, the distribution of countries involved is notably diverse. This diversity likely stems from the significant variations in data sources of individual airlines or airports. Overall, this visualization facilitates a

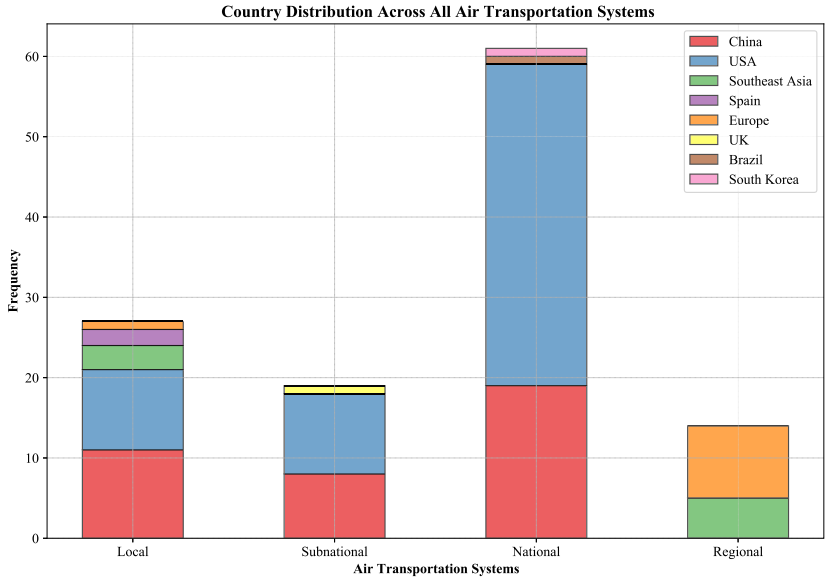



**Fig. 3.** Illustration of country distribution in different air transportation systems.

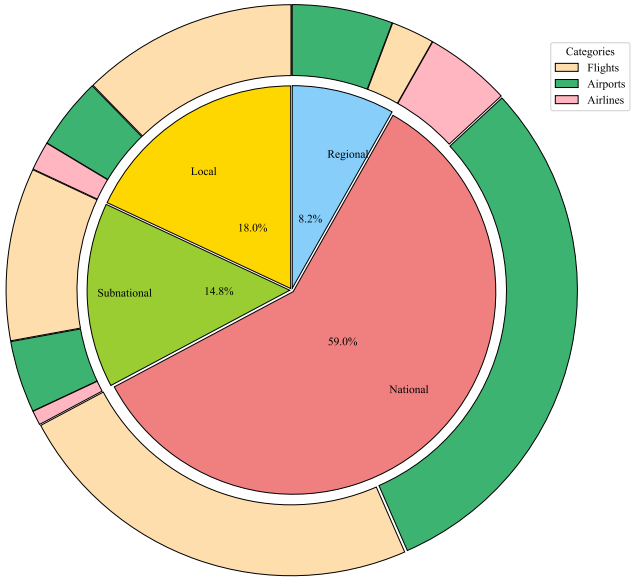



**Fig. 4.** Distribution of references to flights, airports, and airlines in air transportation system research.

comparative analysis of global research efforts in air transportation, underscoring the varying degrees of country-specific engagement within each system.

Fig. 4 illustrates the distribution of references to flights, airports, and airlines within the data descriptions of air transportation system studies. The inner circle reveals a predominant mention of these terms in the national system, indicating a focus on country-level aviation in the data. Conversely, local and subnational systems show a lesser yet notable inclusion of these terms. The outer circle of the figure indicates the thematic prevalence of flights and airports in the national system's data descriptions, while the regional system demonstrates a balanced reference to both, reflecting its emphasis on connectivity. Notably, across all systems, airlines are mentioned less frequently, pointing to a potential area for future research.

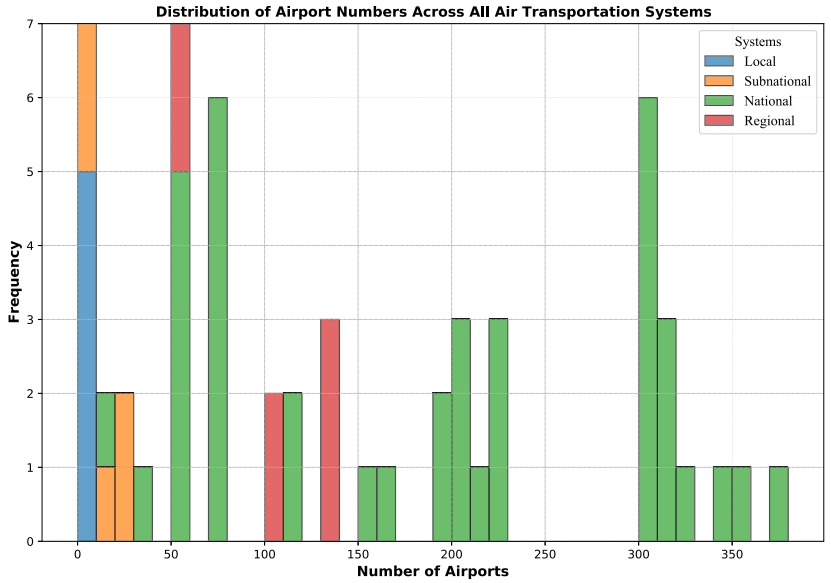



**Fig. 5.** Stratified distribution of airport numbers in different air transportation systems.

Next, we turn our attention to a detailed analysis of the distribution histograms,[2] which depict the frequency of mentions of flights and airports in the data descriptions of studies modeling delay propagation across different air traffic systems. Fig. 5 illustrates the varied distribution of airport numbers in different air transportation systems, highlighting the sector's inherent organizational hierarchy. In the local system, a high frequency of facilities with minimal airport numbers reflects a focus on localized, community-centric air travel. The subnational and regional systems show a moderate increase in airport numbers, signifying their roles as connectors within and across larger areas. The national system, with its diverse range of airport numbers, represents the full spectrum of air travel, from small regional airfields to major international hubs. This gradational distribution underscores the multifaceted nature of air transportation infrastructure, accommodating a wide range of air travel needs.

In Fig. 6, the distribution of flight activity across air transportation systems is clearly delineated. The local system shows a moderate level of flight activity, reflective of regional connectivity and a focus on commuter travel. At the subnational level, there is a noticeable increase in flight frequency, indicating a well-established network that likely supports connections to larger hubs. The national system displays a wide range of flight numbers, emblematic of extensive travel coverage and high-capacity operations at major airports. In contrast, the regional system, though not as extensive as the national, exhibits significant flight activity, underscoring the role of regional carriers and the importance of short-haul flights in domestic aviation. This distribution highlights the hierarchical structure of air transportation systems, ranging from localized operations to comprehensive national networks, each integral to the broader ecosystem of aerial mobility.

### *2.3. Data category and application*

In this section, we are dedicated to demonstrating how various data types are utilized, individually or in combination, to characterize delay propagation behaviors across multiple dimensions. As depicted in Table 2, we categorize the portrayal of delay propagation characteristics into six distinct levels: airport, airline, air transportation networks, multi-airport systems, chained flights, and single flights. Typically, the description of these data at different application levels is used to depict or validate models in experimental sections or to highlight specific types of delay propagation characteristics, each level exhibiting unique features.

Studies focusing on airports or airlines often aim to showcase statistical characteristics of delays at these entities, such as average delay duration, volume or proportion of delayed flights, and their spatio-temporal distribution patterns. Research dedicated to understanding how flight delays propagate through chained flights generally analyzes delay characteristics holistically across different flight legs. Meanwhile, studies from the perspective of air transportation networks employ network metrics – often derived from complex networks – to map delay propagation across the entire airport network. It is important to note that some studies approach this from the perspective of a single flight, characterizing the average delay duration for each flight or examining micro-level flight behaviors through predictive or regression tasks. Multi-airport systems, as the name suggests, are examined from the perspective of a cluster of airports, delineating the delay propagation behavior within this subsystem.

[2] The distribution of airline data description in these systems is not depicted due to the limited number of studies that specifically detail the scale of airlines in our dataset.

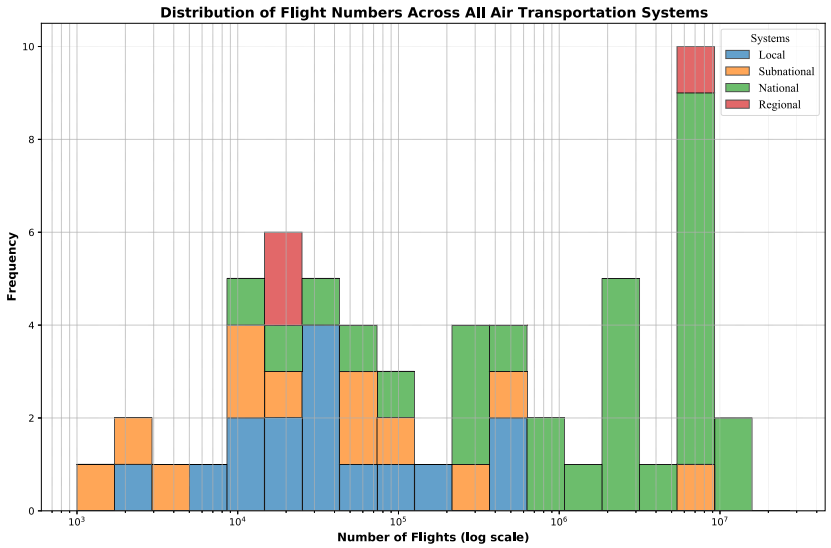



**Fig. 6.** Hierarchical distribution of flight numbers in different air transportation systems.

**Table 2**
Flight delay data categories by application in delay propagation modeling.

| | Sum | Flight operation data | Weather data | Trajectory data | Airspace structure data |
|---|---|---|---|---|---|
| Airport | 48 | 48 | 8 | 7 | 2 |
| Airline | 12 | 12 | 2 | 0 | 0 |
| Air transportation networks | 28 | 28 | 3 | 6 | 1 |
| Multi-airport systems | 1 | 1 | 0 | 0 | 1 |
| Chained flights | 18 | 18 | 2 | 0 | 0 |
| Single flights | 10 | 10 | 3 | 2 | 0 |

Note that the categorization of application scopes within the table is predicated on the characteristics of the cases examined in the experimental sections of the articles, rather than data sources (e.g., originating from airports or airlines) or modeling perspectives (e.g., grounded in flight chain modeling or airport network modeling). Modeling perspectives will be discussed in Section 3. For instance, some studies may utilize flight chain or airport network modeling, yet their analysis primarily focuses on the delay situation from the perspective of a single airport. In these instances, the application scenario is categorized as "airport".
Note that literature without clear data sources or accurate descriptions of the data is not included in the statistics.

Table 2 begins with a column displaying the total number of papers that examine delay propagation across six application scopes. The following four columns detail the frequency with which four different data types are utilized in these scenarios. It is clear that flight operation data is indispensable in flight delay propagation modeling. This data type provides critical field information essential for calculating delays, as evidenced by its application across all scopes. Alongside flight operation data, weather data also plays a pivotal role, given its significant impact on flight delays, making it one of the most frequently used data types. Moreover, flight trajectory data is strategically employed in studies characterizing delay features from the perspectives of airports, air transportation networks, and single flights. In these cases, flight trajectory information often enhances the depiction of operational situations at airports or for individual flights, thereby enriching the delay propagation models. Additionally, airspace structure data proves crucial in modeling delay propagation within multi-airport systems. These systems often share common airspace resources, making the understanding of airspace structure integral to the modeling process. By correlating various types of delay data with their respective applications, we provide a comprehensive reference for data utilization in delay propagation modeling across different levels.

It is important to recognize that different data types do not inherently possess strict advantages or disadvantages; rather, the choice of which data to use primarily depends on the specific requirements of the research question and the integration difficulty of certain data types into the study. For instance, analyses focusing on the performance characteristics of flight delay networks from a medium to long-term perspective rarely incorporate specific operational data such as trajectory or weather information. Similarly, in modeling delay propagation using epidemic spreading models, typically only flight operation data is utilized, as weather information can be challenging to integrate into the model. Moreover, the merits and drawbacks of data are relative, with a hierarchy where flight operation data is primary and other data types like weather and airspace structure data serve as secondary, auxiliary information to consider more operational factors in delay propagation modeling. Nevertheless, it should be noted that the use of any data type depends on the demands of the research question and the characteristics of the chosen methodology.

We then endeavor to provide relevant cases illustrating how different types of data are utilized and combined within the six application levels of research purposes outlined in Table 2. Within the realm of airport-level application research, Belkoura et al.

(2017) leverage trajectory data to identify flights traversing European airspace, then combines this with flight schedule data to calculate each aircraft's delay information and subsequent delays. Carvalho et al. (2016) consider weather condition data as a factor influencing delays and uses association rule mining to reveal whether weather conditions affect flight delays in different regions. Similarly, Sun et al. (2022) employ trajectory and flight operation data to compute flight delays in Europe, incorporating airport capacity as an additional feature into a machine learning model built to predict average airport delays. In research focusing on the operational characteristics of different airlines, Brueckner et al. (2022) incorporate weather information as an additional influencing factor to measure the relationship between the departure delays of preceding flights and the landing delays of subsequent flights. Barnhart et al. (2014) use flight operation data, passenger itinerary data, carrier segment, and ticket data to calculate delay times based on passengers, analyzing the propagation behavior of passenger-centric delays.

In studies aimed at revealing delay propagation behavior from a network perspective, Lin et al. (2021) utilize trajectory data for clustering to identify operational air routes and divide the airspace into grids to construct en-route congestion points. These congestion points are then used as nodes to build a multi-layer air traffic network, which, when combined with flight operation information, is analyzed to understand the network propagation effects of delays. Similarly, Zhang et al. (2021a) and Bao et al. (2021) also integrate weather-type data as additional features into constructed models, forecasting future delay conditions within airport networks. From the perspective of multi-airport systems, Zhao et al. (2022) use airspace structure data to extract key air route points within the system being researched. It then combines this data with flight operation information to calculate delay propagation details associated with these specific route points, providing insight into how delays spread within interconnected airport environments.

In the application context of chained-flight studies, Chen and Li (2019) construct features using flight operation data and weather category data, which are then fed into a random forest model to predict the landing delays of flights. This approach is integrated with a chained flight model to sequentially forecast the delay characteristics of subsequent flights operated by the same aircraft. Wu and Law (2019) employ crew and passenger data to construct a delay propagation network, while also utilizing Bayesian networks to analyze how weather and operational factors influence the delay probabilities of different flights within a flight chain. Finally, from the perspective of single flights, Zheng et al. (2020) utilize weather data to construct features and employs regression analysis to assess its impact on the delay status of individual flights. Similarly, Guo et al. (2020) construct indicators influencing single flight delays by combining weather type data and trajectory information, which are then input into a developed model to predict the delay level of a single flight. In summary, through examples from six application levels of research, we have illustrated how different types of data can be employed for various research questions or objectives. A more detailed description on the methods for modeling flight delays will be elaborated in Section 4.

## 3. Modeling perspectives

Flight delay propagation effects customarily originate from an initial flight delay and subsequently proliferate through flight chains composed of successive flights, ultimately extending to the air transportation networks. As a result, two predominant approaches to modeling flight delay propagation have materialized: flight chain-based models and airport network-based models. These two modeling perspectives analyze delay propagation from micro and macro viewpoints, respectively. By aligning with diverse modeling methods possessing distinct characteristics, these perspectives enable an in-depth and comprehensive investigation of the mechanisms underlying delay propagation.

### 3.1. Flight chain-based modeling

A flight chain delineates a sequence of flights operated by the same aircraft during the day, providing a micro perspective for scrutinizing delay propagation. The notion of a flight chain arises from the propensity for propagated delays to manifest when resources are scarce and interrelated. In scenarios where an inbound flight experiences delays, the likelihood of the outbound flight's delayed departure escalates considerably. Through the flight chain lens, the delay propagation pattern between consecutive flights can be elucidated. Mathematically, the model can be expressed as follows:

$$t_{sd_i} = t_{sa_{i-1}} + T_{turn} + T_{tb} \tag{1}$$

$$t_{ad_i} = \max\left(t_{sd_i}, t_{aa_{i-1}} + T_{\text{turn}} + \varepsilon_1\right) \tag{2}$$

$$t_{sa_i} = t_{sd_i} + T_f + T_b \tag{3}$$

$$t_{aa_i} = t_{ad_{i-1}} + T_f + \varepsilon_2 \tag{4}$$

Here, $t_{sd_i}$ and $t_{sa_i}$ denote the scheduled departure and arrival time of flight $i$. $t_{ad_i}$ and $t_{aa_i}$ denote the actual departure and arrival time of flight $i$. $T_{turn}$ is the minimum turnaround time and $T_f$ is the minimum flying time. $T_b$ and $T_{tb}$ define the flight buffer and turnaround buffer. $\varepsilon_1$ and $\varepsilon_2$ are the random variables signifying the deviation between the minimum flying/turnaround time and the actual flying/turnaround time, which should be positive. We can also employ a simple time–space graph, as presented in Fig. 7, to show the relationship between two sequential flights. In this graph, $sa_i^2$ is the earliest arrival time of flight $i$ from $ad_i$, and $sa_i^3$ is the estimated arrival time of flight $i$ from $ad_i$.

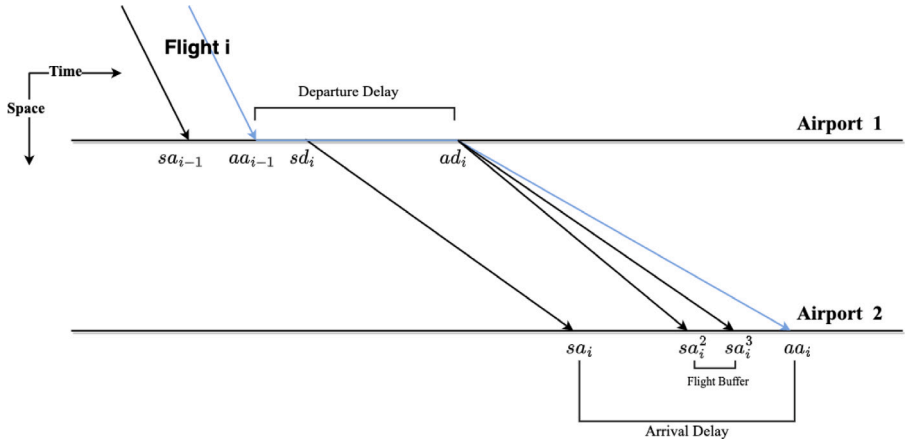



**Fig. 7.** An illustration of the flight chain-based modeling perspective.

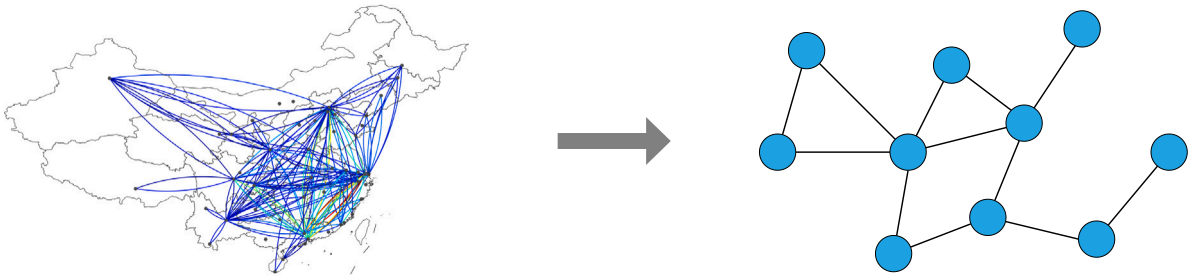


**Fig. 8.** A graphic representation of the airport network-based modeling perspective.

Buffer time denotes the supplementary duration that airlines incorporate into flight schedules and turnaround time, which can effectively mitigate delay propagation. For a more accurate simulation of flight chain dynamics, quantifying buffer time is crucial and necessary. Kafle and Zou (2016) are the first to contemplate the role of buffers in delay absorption and propose a method for buffer time computation. With the buffer time determined, Kafle and Zou (2016) introduce a method to decompose the flight delay into newly formed and propagated delays, which allows for a more comprehensive and intricate examination of delay propagation. Numerous articles have adopted the methodology established by Kafle and Zou (2016), providing in-depth analyses of delay propagation from countries such as South Korea (Kim and Park, 2021) and China (Tan et al., 2021).

In modeling delay propagation from the vantage point of flight chains, various methods can be employed to uncover the patterns of delay propagation. For instance, Chen and Li (2019) enhance the precision of delay propagation prediction for specific flight sequences by merging the merits of machine learning models and flight chain-based modeling. Models grounded in flight chain perspectives can also be integrated with statistical models (Wong and Tsai, 2012; Kim and Bae, 2021), agent-based models (Fleurquin et al., 2013; Cai et al., 2021b), and Bayesian networks (Xu et al., 2005; Wu and Wu, 2018) to analyze the principles governing delay propagation.

### *3.2. Airport network-based modeling*

At a macroscopic level, air transportation systems can generally be conceptualized as networks consisting of airport nodes. Due to the complex and dynamic interactions and interdependencies among these nodes, air transportation networks often manifest the traits of large-scale intricate systems. While delay propagation transpires within local flight chains, the elaborate relationships between airports connected by flights can contribute to the propagation of delays throughout the entire aviation network. As a result, an alternative perspective for modeling delay propagation involves scrutinizing propagation patterns from the standpoint of airport networks. We present the generic modeling representation that centers on airport networks in Fig. 8.

In this context, airports can be regarded as nodes, and the intricate connection relationships between these nodes afford numerous potential definitions for edges. For instance, edges may be defined based on distance (Zeng et al., 2021), flow volume (Wang et al., 2020b), and delay relationships (Mazzarisi et al., 2020). Several studies also investigate the causal relations between airport nodes, constructing causal delay networks via the Granger causality test and incorporating complex network analysis to assess pertinent performance indicators of the network (Zanin et al., 2017; Du et al., 2018). Moreover, in network representation methodologies, complex networks, Bayesian networks, and graph neural network approaches primarily rely on airport network-based models, analyzing the patterns and features of network delay propagation founded on this premise.

## 4. Modeling methods

In our examination of flight delay propagation, we explore a range of modeling methods,[3] each with its unique focus and application. These include economic models, statistical models, machine learning models, epidemic spreading models, simulation-based models (further divided into agent-based and general simulation-based models), queueing models, and network representation models, which encompass Bayesian networks, complex networks, graph neural networks, and alternative spatio-temporal methods. These methodologies provide a foundation for understanding the principles and characteristics of flight delay propagation. In subsequent chapters, we discuss how these various approaches can be complemented and enhanced by advanced technologies to achieve a more nuanced analysis of delay propagation features.

### 4.1. Economic models

Economic models provide an efficient way to describe and analyze the impact of delay propagation within air traffic networks. After an extensive review of previous papers, we find that statistical models share many similarities with economic models, and certain econometric methods are efficient for significance analysis. We have laid down several criteria to distinguish these models. The distinct characteristics of economic models, as outlined in the paper, include:

(1) the introduction of various indicators pertinent to delays;
(2) the occasional employment of statistical methods, such as regression analysis, for validation or estimation purposes, albeit with a foundational focus on describing or explaining the genesis and spread of delays based on certain principles or theoretical underpinnings[4];
(3) the exploration of delay propagation relationships between successive flights by models.

The first category of study introduces various indices to represent distinct aspects and characteristics since many of them are not straightforward. These indices more directly reflect the nuances of delay propagation (Beatty et al., 1999). Proceeding from reality and common knowledge, the second emphasizes identifying and analyzing factors that influence delay propagation at various levels (Wang et al., 2003; Kafle and Zou, 2016). The third category concentrates on exploring solutions for delay propagation across successive flights, examining the methods and strategies employed to mitigate such delays (Brueckner et al., 2021b).

The Delay Multiplier (DM) is the earliest proposed index to describe the impact of delays on flight schedules. It measures the ratio of the initial delay (I) and the summation of the downstream delay(D), namely the total propagated delay (Beatty et al., 1999).

$$DM = \frac{I + D}{I} \tag{5}$$

This index offers a lucid and efficient representation of the propagated effect of an initial delay. The original study employs a delay tree methodology alongside various regression models for index calibration. However, its limited detail and potential ambiguity in distinguishing between different types of delays render it somewhat inadequate in addressing the costs of flight cancellations, a gap that subsequent research has sought to bridge. The subsequent adaptations of the Delay Multiplier (DM) index have been notably advanced by Welman et al. (2010), who introduce a preliminary method for classifying delays at individual airports. This approach effectively highlights the differences in delay propagation and offers a pragmatic framework for incorporating the benefits of reduced delay propagation into decision-making processes for airport investments. Belkoura et al. (2017) offer the first modification of the DM index, basing the reformulation on a nonlinear relationship between average inbound and outbound delays at an airport within a one-hour time window. The negative normalized quadratic coefficient provides additional information that outbound delays are not more driven by inbound ones for large delay values. Considering the limitations of previous indices, such as the DM index, in the context of single airports, Zhang et al. (2019) establish an innovative delay Propagation Index (PI) to capture the structural dependency caused by the delays propagated between pairwise airports. Defined as the statistically significant transfer entropy obtained by pairwise test, PI is incorporated into a neural network regression model to estimate the average departure delay. To complement this, Wu and Law (2019) develop two advanced metrics that incorporate the posterior delay distribution, known as the expected delay multiplier $DM_e$ and expected propagated delay multiplier $DM_x$.

The efficiency of economic models also hinges on employing suitable econometric methods to scrutinize factors associated with delay propagation and their significance. Wang et al. (2003) are the first to distinguish controllable and variable factors in delay propagation calculations, introducing better and simpler delay propagation equations for both departure and arrival. Hao et al. (2014) utilize a different econometric model to assess how delays at New York (NY) airports influence the broader National Airspace System (NAS) airports. Their approach involves separate considerations for airspace and each major airport region. By applying two simultaneous models and employing the two-stage least squares regression method, they analyze the delays of NY airports and NAS airports. Their findings predominantly attribute the congestion at New York airports to the flight traffic specific to New York. Tan

[3] Our focus is primarily on the methodologies employed, rather than making a distinction between theoretical and empirical research. This approach is rooted in our observation that a significant proportion of the literature on flight delay propagation research tends to blend theoretical frameworks with empirical data. As a result, our classification is geared towards elucidating the nature and mechanisms of the modeling methods, recognizing that these approaches frequently integrate theoretical concepts with empirical validation.

[4] Unlike statistical models, which primarily focus on data analysis, economic models' regression techniques are more oriented towards elucidating the underlying causes or genesis of observed phenomena.

et al. (2021) propose an economic regression model to analyze delay propagation patterns using data from Chinese airlines. Their findings reveal a distinct difference from delay propagation observed in European and US airlines, indicating that in China, delay propagation is not linked to poor performance of on-time performance (OTP). Moreover, they identify that hub airports and low-cost carriers contribute differently to the dynamics of delay propagation.

More precisely, Kafle and Zou (2016) strictly distinguish the function of buffer in absorbing propagated and newly formed delays. For three scenarios, newly formed delays absorbed first, the propagated delay absorbed first, or both of them reduced simultaneously, leading to different algorithms that calculate the total propagated delay. To accurately model the inception of propagated delays and circumvent biases inherent in straightforward ordinary least squares analyses, Kafle and Zou (2016) employ an econometric approach incorporating a latent variable framework based on a discrete choice model. Subsequently, they perform regression analysis on the propagated delay, utilizing three distinct groups of variables, applying a two-step procedure or full information maximum likelihood (FIML) method to estimate the coefficients of the regression.

More recently, Brueckner et al. (2021a) explore the contribution of ground buffer and flight buffer to eliminate the propagation delay. Their research involves the development of a two-flight model, an extension of the single-flight model, wherein they rigorously define the cost implications for consumers. Through meticulous mathematical analysis, Brueckner et al. (2021a) establish that the flight buffer of the first flight is inconsequential to the propagation of delay, underscored by a thorough examination of varying coefficient conditions. In a subsequent paper, Brueckner et al. (2021b) in another article have provided a simpler version two-flight model, using certain probability to replace the probability density function in the previous model, thereby enabling a more theoretical exploration of the contributions of different buffers. This study also outlines several prospective avenues for future research in this domain. Most importantly, this study is unique in that it exclusively incorporates theoretical analysis without empirical support. To further highlight the distinct analytical approaches employed in our study, we have introduced an additional column in the table that follows, which is named "Analysis Category", which is specifically designed to categorize and clearly distinguish the types of analysis – namely, theoretical, empirical, or a combination of both – utilized in the investigation of flight delay propagation and air transportation management. Advancing the field further, Brueckner et al. (2022) examine the direct correlation between the propagated departure delay and the arrival delay of an inbound flight. Their analysis focused on individual flights and their immediate predecessors, incorporating route-specific characteristics. These studies underscore the significance of utilizing appropriate econometric methods for a deeper understanding of the variables affecting delay propagation and their relevance in various contexts. Comprehensive findings from these studies are presented in Table 3.

Econometric models for studying flight delay propagation are categorized into three distinct approaches, each with unique characteristics and applications. The first introduces new indices to quantify delay propagation, offering clarity in distinguishing between propagated and new delays, though its exploration of influencing factors is somewhat narrow, often limited to variables like airport type and airline. The second utilizes linear regression to highlight specific variables' contributions to delays, providing broad applicability and practical insights, especially valuable for commercial use due to its multi-dimensional analysis. The third, characterized by a rigorous mathematical approach, delves deeply into single and two-flight models, offering detailed insights but with limited applicability due to its complexity. While this approach is academically valuable, particularly for detailed case studies, its specificity may restrict widespread use. Each category presents a balance of strengths and limitations, guiding their selection based on the research focus and the scope of application.

### *4.2. Statistical models*

Instead of emphasizing understanding the underlying mechanisms through regression or focusing more on the explanation of the models themselves like economic models, the pivotal role of statistical models in the realm of flight delay and delay propagation analysis is twofold: firstly, they facilitate the quantification of the significance of various influential factors; secondly, they aid in uncovering patterns in delay propagation through the lens of statistical distributions and prevalent statistical models. We sort the models into statistical models as below principles:

(1) The primary emphasis is on the interrelationship among data points, exemplified by the use of regression analysis to probe into the factors influencing delays.
(2) These models are utilized to characterize statistical patterns or trends in delays, as opposed to elucidating their underlying causes based on theoretical principles or assumptions.

The first type of study employs standard statistical methods to investigate data characteristics and elucidate the mechanisms of delay propagation (Wong and Tsai, 2012). On the other aspect, the second type utilizes common distribution models to accurately fit and describe the patterns of delay distributions (Cao et al., 2019). Table 4 in the document offers a comprehensive overview of pertinent studies and summarizes the application of statistical models in the analysis of flight delays and their propagation.

Tu et al. (2008) concentrate on identifying key factors that influence the distribution of departure delays and on developing a predictive model that quantifies daily and seasonal trends using non-parametric methods. In their approach, a mixed distribution is initially employed to estimate residual errors and a modified expectation–maximization (EM) algorithm is developed to avoid getting trapped in local optima. Nayak and Zhang (2011) examine how delays at a single airport can affect the entire system. They use a multivariate simultaneous model, which takes into account the cumulative factors causing delays. Their study emphasizes the bidirectional relationships between network nodes and concluded that queuing delays and extreme weather conditions are the primary contributors to these disruptions. Carvalho et al. (2016) advance data mining techniques through the use of data indexing and the analysis of association rules, enabling them to compute confidence and support ratios. This method facilitates the discovery of

**Table 3**
Summary of the characteristics and application scope of economic models.

| Literature | Target | Analysis category | Metrics/Methods | Description | Application |
|---|---|---|---|---|---|
| Beatty et al. (1999) | Quantify and evaluate the delay propagation | Both | Delay Multiplier (DM) | The ratio of the sum of initial delay and downstream delay of initial delay | Chained flight delay |
| Welman et al. (2010) | Calculate delay propagation multipliers for cost–benefit analysis | Both | Composite Airport Multipliers | Calculate total original delays and propagated delays for a combination of airports | Airport delay |
| Kafle and Zou (2016) | Understand delay propagation patterns and associated mitigation measures | Both | Total propagated delay (TPD) | Quantify the newly formed and propagated delay and analyze how buffer absorbs these delays under three scenarios | Airport, airline, and chained flight delay |
| Belkoura et al. (2017) | Extract a more complete relationship between the inbound and outbound delays | Both | A polynomial fit between inbound and outbound average delays | Synthesis standard delay multiplier and additional response triggered by abnormal delays | Airport delay |
| Zhang et al. (2019) | Analyze the delay propagation effect among pairs of airports | Both | Propagation index (PI) | Causal interrelationship based on transfer entropy | Airport delay |
| Wu and Law (2019) | Model the delay propagation effects via multi-connected airline networks | Both | Expected delay multiplier $DM_e$ and expected propagated delay multiplier $DM_x$ | Incorporate the stochastic factors influencing delay distribution | Chained flight delay |
| Wang et al. (2003) | Explore the fixed and variable factors that characterize delay propagation | Both | Econometric regression | Separate the influential factors and put forward a recursive delay propagation model | Airport and chained flight delay |
| Hao et al. (2014) | Evaluate how much delays at the three New York airports contribute to the national airspace system delay | Both | Econometric regression | Consider various influential factors and scenarios | Airport delay |
| Tan et al. (2021) | Quantify and depict the delay propagation pattern in Chinese airlines and airports | Both | Econometric regression | An exploratory analysis based on econometric regression | Airport and airline delay |
| Brueckner et al. (2021a) | Explore the contribution of ground buffer and flight buffer to eliminate the propagation delay | Both | Econometric regression | An exploratory analysis based on econometric regression | Airline delay |
| Brueckner et al. (2022) | Identify propagated delays and measure how they contribute to the arrival delays | Both | Econometric regression | A simple method which considers the individual flight and its immediate predecessor | Airline delay |
| Brueckner et al. (2021b) | Revisit the buffer-choice problem in a simpler model where the random shocks influencing flight times are discrete | Theoretical | Simple-flight model and two-flight model | Fully theoretical analysis on the contribution of different buffers | None |

We have added a new column named "Analysis category" with three possible values: "Theoretical", "Empirical", and "Both", to illustrate the analysis category of this study. However, the studies we find only conclude "Both" and "Theoretical".

connections between sequential delays. Notably, they reveal that adverse meteorological conditions and departure times significantly influence delays. Furthermore, their research underscores a strong correlation between departure and arrival delays. Zheng et al. (2020) apply the ordinary least squares and quantile regression to investigate the impact of various influencing factors, which are categorized into two groups according to whether the influencing factors are related to delayed propagation or flight departure delay.

Survival analysis is a popular method for quantifying flight delay and delay propagation, as it allows airlines to examine the chances of recovering from flight delays through the calculation of hazard ratios. Wong and Tsai (2012) are among the pioneers of using survival analysis to study the patterns of flight delay and its propagation. Specifically, they apply the Cox proportional hazards model to investigate flight delay propagation along flight chains. Additionally, they apply the Kaplan–Meier estimator and log-rank method to examine the potential impact of factors on departure and arrival delays. Based on the same methods, Kim and Bae (2021) conduct a similar analysis focusing on flight delays in South Korea and change some factors.

Comparative analysis, a research methodology employed to compare and contrast various elements, cases, or methods to draw conclusions or identify patterns and relationships, is also extensively utilized in discerning delay propagation patterns. Wu et al. (2018b) employ and compare various copula functions to ascertain the statistical distribution of delay sequences and analyze

**Table 4**
Summary of the characteristics and application scope of statistical models.

| Literature | Target | Factors/Distribution | Method | Application |
|---|---|---|---|---|
| Tu et al. (2008) | Estimate departure delay distributions | Daily, seasonal trends and residual errors | Non-parametric methods and mixture distribution | Single flight delay |
| Nayak and Zhang (2011) | Measure the significance of influential factors | Average daily arrival delay, deterministic queueing delay, weather patterns, aircraft equipment type | Multivariate simultaneous regression models | Airport delay |
| Wong and Tsai (2012) | Capture the dynamic characteristics of flight delay | Buffer time, aircraft type, weather, passenger and cargo handling | A cox proportional hazards model | Single flight delay |
| Carvalho et al. (2016) | Unveil hidden patterns of flight delays | Adverse meteorological conditions | Data indexing techniques and the association rule | Airport and airline delay |
| Wu et al. (2018b) | Investigate the correlation between sequential flight delays caused by other factors | Airlines, weather, and ATC | The copula function | Single flight delay |
| Zheng et al. (2020) | Investigate the impact of various influencing factors on flight departure delay | Delay propagation-related factors and other factors | Ordinary least square and quantile regression models | Single flight delay |
| Lykou et al. (2020) | Assess the risk of delay incidents in airports and how the risk propagates in the aviation network | Delay propagation-related factors and risk analysis | Build a risk-based dependency graph | Airport and chained flight delay |
| Kim and Bae (2021) | Evaluate the actual impacts of major variables on flight delays | The turnaround buffer | Kaplan–Meier estimator, log-rank test, and Cox proportional hazards model | Airport delay |
| Cao et al. (2019) | Explore the internal mechanism of flight departure delay | Power-law distribution | Statistical analysis and summation | Airport delay |
| Wang et al. (2020a) | Investigate the mechanisms of propagated delays and evaluate the airline's operational effectiveness | A shifted power law or an exponentially truncated shifted power law delay distribution | The complementary cumulative distribution functions | Airline delay |
| Mitsokapas et al. (2021) | Illustrate flight delay distribution | Power-law decay distribution | A maximum likelihood estimation method | Airport delay |
| Wang et al. (2022b) | Identify different timescales of delay interactions | Stable time lags | A time delay stability (TDS) algorithm | Air transportation network delay |

the propagation effect from the perspective of interdependency among sequences of flight delays. Lykou et al. (2020) focus on the variable performance and stochastic nature of flight networks. They utilize a risk-based approach to clarify the intricate interdependencies and resultant delays in aviation networks. This is accomplished by creating airport dependency graphs and calculating indices that concisely encapsulate delay and dependency factors. Additionally, the authors propose two normalization methods: the min–max method and the standard method. Both are designed to quantify the effects of flight delays on passenger satisfaction and airport operational efficiency, enhancing understanding of the impact of delays.

Recent studies have increasingly utilized combined tools to evaluate features and construct flight delay patterns. These studies aim to provide a more comprehensive understanding of the underlying patterns and factors associated with flight delays and their propagation. Cao et al. (2019) divide all factors into propagation factors and non-propagation factors (NPF). They develop a specific statistical measure for formulating flight planning, finding that the distribution of flight departure delays caused solely by NPFs can be explained by a queueing model, showing a power law (PL) distribution, while the original distribution of flight departure delay follows the shifted power law (SPL). Wang et al. (2020a) introduce a universal metric to evaluate the operational effectiveness in delay alleviation by constructing delay propagation with dynamic models. They group all airlines into either an SPL or an exponentially truncated shifted power law delay distribution group. A similar thought is presented by Mitsokapas et al. (2021), which divides the distribution at peak point and uses power law and exponential function to model the probability density function of arrival delays, aiming to explore the delay propagation and other factors' impact. To illustrate the timescales of delay propagation, Wang et al. (2022b) further apply a time delay stability (TDS) algorithm for their designed delay stability network and delay lag network, which separately indicates the stable lag and the actual magnitude of lags. This approach helps to understand the dynamics of delay propagation across different time scales.

Statistical models for flight delay propagation are divided into two main approaches, each tailored to specific analytical needs. The first employs classical statistical models to assess the wide-ranging impacts of various factors on flight delays, surpassing econometric models in breadth by evaluating the influence of airports and airlines on the entire aviation network. This approach

**Table 5**
Summary of the characteristics and description of machine learning models.

| Literature | Modeling perspective | Data | Method | Description |
|---|---|---|---|---|
| Rebollo and Balakrishnan (2014) | Airport network | ASPM data | K-means clustering + Random forests | Identify network delay states with k-means clustering and apply random forests for delay classification and prediction |
| Chen and Li (2019) | Flight chain | BTS data + NOAA data + ASPM data | Delay propagation model + Random forests | Predict arrival delay and compute late arriving aircraft delay iteratively |
| Yu et al. (2019) | Flight chain | PEK airport data | DBN + LSTM | Develop DBN to extract deep delay patterns and SVR to perform supervised fine-tuning |
| Güvercin et al. (2020) | Airport network | Research and Innovative Technology Administration (RITA) | CAM model + REG-ARIMA | Utilize the clustering method to represent delay time series and employ a network metric to predict airport delay |
| Sun et al. (2022) | Airport network | Eurocontrol R&D data archive | Recurrent neural networks | Apply random forests to select features for flight prediction and LSTM to predict airport average delay |

excels in identifying dependencies within the network, facilitating the development of targeted mitigation strategies essential for complex air traffic system management. However, its effectiveness wanes when confronting complex, non-linear relationships, a notable weakness that limits its applicability in more intricate scenarios. The second category concentrates on delay distributions, using exponential functions for advanced predictions based on observed data. This method, more nuanced than linear regression, is crucial for proactive delay mitigation strategies but, like the first, may struggle with the direct causation analysis of delays due to its focus on probabilistic outcomes rather than clear causal pathways. Together, these statistical methods offer valuable insights into understanding and predicting flight delays, with their strengths in broad network analysis and predictive modeling somewhat counterbalanced by their limitations in dissecting complex, non-linear interactions within the aviation ecosystem.

### *4.3. Machine learning models*

Machine learning models for delay propagation prediction are designed to forecast plausible delay values within a specified time frame, with the objective of minimizing discrepancies between predicted and actual values, thereby enhancing overall prediction accuracy and effectiveness. Indeed, machine learning algorithms have seen extensive applications in flight delay prediction tasks in recent years, particularly in the areas of delay classification and forecasting for individual flights (Carvalho et al., 2021). However, as this review concentrates on research related to flight delay propagation modeling, the focus is placed on examining studies that employ machine learning algorithms specifically for analyzing delay propagation or network effects within the aviation industry. The pertinent literature, along with their summarized findings, can be found in Table 5.

In the study of flight delay propagation, machine learning methods are predominantly employed for predicting flight delays and identifying potential contributing factors within the constructed aviation networks. These models adeptly handle large volumes of delay data, learning complex combinations of features. Notably, specific models like random forests are utilized to forecast the network's delay status (Rebollo and Balakrishnan, 2014). Moreover, machine learning approaches can be integrated with flight chain-based modeling techniques (Chen and Li, 2019). This integration allows for the prediction of individual flight delays, which are then inputted into the flight chain model to iteratively output the characteristics of delay propagation in successive flights.

Rebollo and Balakrishnan (2014) present a novel approach that combines the K-means clustering algorithm with the Random Forest method to construct temporal and spatial explanatory variables. This integration facilitates the classification and prediction of future delay scenarios for specific origin–destination (O–D) pairs within the aviation network, enhancing the overall understanding and management of delay propagation dynamics. Yu et al. (2019) introduce an innovative framework that combines deep belief network (DBN) and support vector regression (SVR) to uncover the intrinsic patterns of flight delays and identify the micro-level key factors influencing delays. To make chained flight delay prediction, Chen and Li (2019) propose a hybrid model that combines the strengths of the flight chain-based model and machine learning methods, enabling the analysis of large-scale datasets while capturing the intrinsic relationships between airports. Furthermore, feature selection is employed to improve prediction accuracy.

From the network perspectives, Güvercin et al. (2020) apply a model called "Clustering airport modeling" (CAM) to represent airport delay series. They utilize network metrics such as betweenness centrality to predict flight delays based on the fitting performance. Sun et al. (2022) develop different types of machine learning models, including a random forest model and long short-term memory (LSTM), for predicting flight delay propagation and analyzing airport network dynamics.

Machine learning methods in modeling flight delay propagation are distinguished by their ability to analyze complex datasets, efficiently identify non-linear patterns, and forecast future delays. While these techniques excel in processing vast and intricate data, their application often requires expert knowledge for manual feature construction, alongside significant computational resources. This necessity for specialized expertise in feature engineering highlights a critical aspect of their implementation, yet their effectiveness in revealing the underlying dynamics of flight delays underscores their value in the field.

### 4.4. Epidemic spreading models

Epidemic spreading paradigms constitute mathematical frameworks devised to emulate the infectious propagation within a population. These models segregate individuals into discrete states and facilitate transitions between these states in accordance with specific infectious schemes. Specifically, the Susceptible-Exposed-Infectious-Recovered (SEIR) model is a widely used compartmental model in epidemiology, which divides the population into four compartments. Distinct state designations correspond to diverse populations, which, in turn, display varying attributes in relation to disease susceptibility and transmission. Consequently, the delineation of an extensive array of states and transition mechanisms can proficiently simulate the disease dissemination process. Moreover, utilizing epidemic dissemination paradigms, such as the SEIR model, for the analysis of flight delay propagation necessitates the establishment of parallels between the dynamics of disease transmission and the proliferation of delays throughout the air transportation system. The SEIR model can be represented as the following system of ordinary differential equations (ODEs):

$$\begin{aligned} \frac{dS}{dt} &= -\beta \frac{SI}{N}, \\ \frac{dE}{dt} &= \beta \frac{SI}{N} - \sigma E, \\ \frac{dI}{dt} &= \sigma E - \gamma I, \\ \frac{dR}{dt} &= \gamma I. \end{aligned} \tag{6}$$

where $\beta$ and $\gamma$ denote the transmission rate and recovery rate respectively. $N$ is the total population size and $\sigma$ is the rate at which exposed individuals become infected. Besides, the variables $S$, $E$, $I$, and $R$ represent the number of individuals in the Susceptible, Exposed, Infected, and Recovered states, respectively. These categories are pivotal in modeling the dynamics of disease spread within a population.

Epidemic spreading models in flight delay propagation draw a parallel with the spread of infectious diseases, viewing the process through the lens of systems dynamics (Baspinar and Koyuncu, 2016). Initially, localized disruptive events at airports or in airspaces can lead to delays in individual flights. These flights, in turn, have the potential to propagate this effect to other airspaces and airports at specific rates. These rates can be quantified as certain proportional parameters with physical significance, such as the flow ratio between two airports, which can be calculated using historical data. Furthermore, these models establish a set of state equations for network delay propagation, characterizing the behavior of flight delay spread. These states are typically associated with delays and are often formulated as a system of ODEs. To elucidate the evolution of network delay propagation, these ODEs can be solved using methods like Euler's method or transformed into a system of difference equations. The solutions to these equations provide insights into the unknown parameters and the evolving patterns of network delay propagation.

In this context, airports or flights can be treated as nodes in a network, with the delay propagation acting as the "infection" that spreads among them. To this end, approaches to modeling flight delay propagation via epidemic spreading models can be broadly categorized into two distinct methodologies: airport-based and flight-based methods (Baspinar and Koyuncu, 2016). In their work, the flight-based epidemic model bifurcates flights into susceptible and infected states. Conversely, the airport-based epidemic model characterizes the entire airport as a metapopulation, thus adopting a distinct approach to modeling disease transmission dynamics. Wu et al. (2019) propose a flight-based Susceptible-Infected-Susceptible (FSIS) model to examine the flight delay process within air transportation networks and discern factors that serve to either exacerbate or mitigate the delay propagation. Besides, airport-based models frequently incorporate complex network theory, manifesting in various network configurations as demonstrated by Zhang et al. (2020). Similarly, Dai et al. (2018) formulate a novel congestion propagation model for departure aircraft within a multi-event and multi-stage scheduling framework, grounded in the principles of heterogeneous flight connections and network construction. Li et al. (2020) elucidate the macroscopic manifestation of delay propagation and long-term evolution within airport networks by employing the Susceptible-Infected-Recovered-Susceptible (SIRS) epidemic model. Ceria et al. (2021) utilize a heterogeneous SIS spreading process on an airline network to model airport congestion contagion, with the overarching goal of replicating airport vulnerability. Within this framework, the heterogeneous infection rate and recovery rate are evaluated based on the constructed network. The related studies and their summaries are listed in Table 6.

In the realm of parameter estimation, Baspinar and Koyuncu (2016) adopt a data-driven methodology for ascertaining parameter values. This approach entails the computation of transmission rates by leveraging historical flight data and the development of an ODE-solving algorithm to estimate recovery rates. Additionally, there are some econometric methods used to estimate transmission rates, such as the translog model (Wu et al., 2019) and factor analysis (Zhang et al., 2020). Moreover, data-driven approaches may be amalgamated with complex network metrics to facilitate a more precise estimation of network epidemic transmission and recovery rates (Ceria et al., 2021).

Epidemic spreading models, applied to flight delay propagation, draw on epidemiology principles to analogize delay spread to disease transmission, effectively capturing the systemic patterns of delay propagation across the network. This approach excels in illustrating how delays can spread through an air traffic system, mirroring the spread of an epidemic. However, the model faces challenges in delineating the specific states of delay transmission and may oversimplify the causes of individual delays. The variability in propagation processes presents difficulties in modeling based on standard epidemiological equations. Despite these limitations, epidemic spreading models are particularly suited for analyzing network-wide delay spread, offering valuable insights into the overarching mechanisms of delay propagation within the aviation system.

**Table 6**
Summary of the characteristics and classification of epidemic spreading models.

| Literature | Data | Parameter estimation | Model type | | Epidemic models | | |
|---|---|---|---|---|---|---|---|
| | | | Flight based | Airport based | SIS | SIR | SIRS |
| Baspinar and Koyuncu (2016) | European airports data | Data-driven method | ✓ | ✓ | ✓ | | |
| Dai et al. (2018) | ATL airport data | Data-driven method | | ✓ | | ✓ | |
| Wu et al. (2019) | Airline data | Translog model | ✓ | | ✓ | | |
| Li et al. (2020) | CAAC data | Numerical simulation | | ✓ | | | ✓ |
| Zhang et al. (2020) | Simulation data | Factor analysis | | ✓ | | ✓ | |
| Ceria et al. (2021) | BTS data | Network metrics analysis | | ✓ | ✓ | | |

### 4.5. Simulation-based models

A simulation model represents a parameterized framework that necessitates computational resolution, as its complexity precludes analytical solutions. Given the dynamic, mixed, and highly nonlinear nature of air transportation systems, accurately modeling propagation dynamics commonly proves formidable. As an alternative, simulation-based models can be employed, primarily bifurcating into agent-based models and general simulation-based models.

#### 4.5.1. Agent-based models

Agent-based models serve as a potent instrument for representing distributed complex systems. This approach substantially enhances the analysis and simulation of flight delay propagation by facilitating the system's organic abstraction and decomposition into numerous agents operating concurrently, engaging in mutual interactions, and coexisting within a dynamic environment (Chen and Cheng, 2010). In an agent-based model, agents adhere to a predefined set of rules established by the users, which guide them towards cooperatively achieving a global objective. Consequently, this engenders intricate behaviors on a macroscopic scale (Qin et al., 2019). The agent-based modeling approach is structured as follows:

(1) Abstract the airport network, determine what constitutes agents (which can be flights, airports, etc.), and define the interactions between agents.
(2) Implement the model experiments and analyze the results for specific research questions.

In recent years, the agent-based method has been broadly implemented in flight delay propagation. The perspectives adopted in relevant literature can be classified into two primary categories:

- Predict the delay process by simulating the delay propagation in aviation networks (Ciruelos et al., 2015; Campanelli et al., 2016; Guleria et al., 2019; Cai et al., 2021b; Wang et al., 2021).
- Analysis of several factors, such as flight rotation, flight connectivity, and airport congestion, on the impact of delay propagation (Fleurquin et al., 2013; Qin et al., 2019).

Such network modeling approaches prove efficacious and cost-effective in application, as they can replicate nominal conditions, reproduce extreme case scenarios, and facilitate "what-if" case studies (Ciruelos et al., 2015). Campanelli et al. (2016) utilize two distinct agent-based models to simulate delay propagation and evaluate the ramifications of network disruptions across various regions. Subsequently, the researchers compare the efficiency of a first-come, first-served protocol with an Air Traffic Flow Management (ATFM) slots priority system.

As the model's abstraction level exerts a direct influence on computational efficiency and simulation accuracy, the agent configuration and parameter consideration become essential. Fig. 9 showcases the architecture of delay propagation modeling by employing agent-based models. The main components of the agent-based model built for flight delay propagation can be summarized as follows:

***Agent:*** An agent denotes an entity capable of functioning in alignment with its designed objectives, perceiving alterations in its environment, and engaging in communication with fellow agents (Chen and Cheng, 2010) . In the model of delay propagation, each flight is represented as an agent. Each agent needs to: (1) acquire the flight schedule from the environment; (2) fulfill the journey from departure at the origin to arrival at the destination; (3) receive information from the environment and interact with other agents to update its state, which includes turnaround, departing, flying, arriving, and so forth.

In some works, each airport is modeled as an agent due to the careful consideration of the airport's operational state, such as the degree of saturation (Wang et al., 2021). The airport agent should: (1) obtain the information of flights; (2) update its state in accordance with capacity.

***Environment:*** The agents' environment is characterized by dynamism, diversity, and finite resources (Guleria et al., 2019). Within the delay propagation model, the environment pertains to the entire aviation network, potentially encompassing information regarding weather, crew, and routes. All agents operate in a structured manner within this environment.

***Interaction:*** Interaction implies that agents are influenced by other agents and adhere to the rules established by users in order to pursue their objectives and fulfill their responsibilities (Guleria et al., 2019). The interaction of a flight agent can be divided into three parts: (1) interaction with its previous flight; (2) interaction in the flight duration; (3) interaction with the airport.

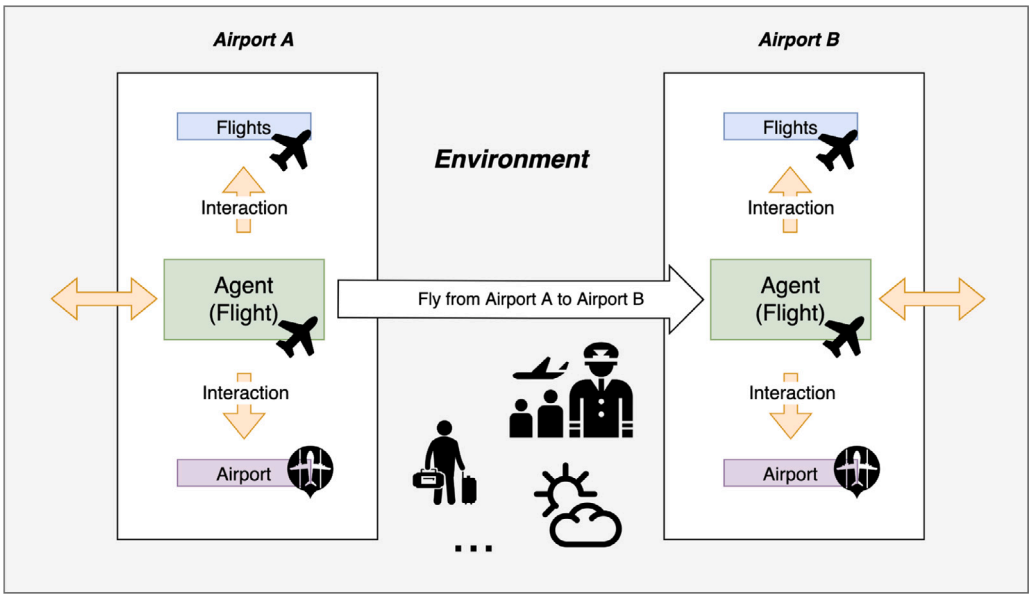



**Fig. 9.** An illustration of agent-based models.

*Interaction with its previous flight:* The delay experienced by an upstream flight can substantially impact the delay of downstream flights due to flight connections. The relationship between the successive flights can be described as Eqs. (1), (2) (Guleria et al., 2019)

Regarding the specific modeling, Qin et al. (2019) consider the likelihood of early flight departure and establish a 5-min threshold for early departure. Wang et al. (2021) adopt a conditional probability model to predict the expected departure time. In addition to examining the delay propagation resulting from aircraft, several papers also investigate the impacts of crew and passenger connectivity (Fleurquin et al., 2013; Qin et al., 2019). The findings indicate that aircraft rotation and connectivity of crew and passengers exert a more substantial influence on the outcomes of delay propagation modeling (Fleurquin et al., 2013). Additionally, existing literature rarely takes the ground buffer into account in the modeling process.

*Interaction in the flight duration:* The interaction in the flight duration is considered as Eqs. (3), (4). Airlines frequently incorporate buffer time into flight durations to reduce the incidence of arrival delays, and departure-delayed flights may accelerate during flight time (Qin et al., 2019). Factors such as weather can also influence flight time. To accommodate all these factors, interactions in flight duration are represented by estimating actual flight time, with machine learning methods being preferred. Cai et al. (2021b) adopt multivariate nonlinear regression by utilizing the historical flight operation data. Wang et al. (2021) develop a random forest regression model to account for additional factors, such as aircraft type and date. It is noteworthy that some models (Fleurquin et al., 2013; Guleria et al., 2019) make the simplifying assumption that delays cannot accumulate in the air. Consequently, these models directly substitute scheduled flight time for actual flight time.

*Interaction with the airport:* Airport capacity for both departure and arrival is constrained within specific timeframes. When a flight agent requires takeoff or landing clearance, it must interact with the target airport to establish an appropriate departure or arrival sequence. Flight agents can proceed without queueing if the demand remains at or below the airport's capacity. However, when demand surpasses capacity, queueing strategies are employed to coordinate the orderly takeoff and landing of multiple aircraft. These strategies encompass the "first come, first served" principle (Guleria et al., 2019) and the priority-based strategy, which accounts for economic losses (Qin et al., 2019). Other approaches, such as random forest regression, have also proven effective (Wang et al., 2021). Besides, Ciruelos et al. (2015) contemplate the inclusion of various external perturbations in their modeling endeavors. In Table 7, a compilation of all 7 associated articles utilizing agent-based models, along with their respective summaries, can be found.

Agent-based models adeptly capture the micro-level interactions within air traffic systems, delivering intricate insights into individual behaviors and specific operational scenarios. This approach excels in detailed analysis, revealing how each entity's actions influence overall delay propagation. However, their depth comes at the cost of being computationally intensive and requiring substantial data. Ideal for dissecting complex operational scenarios, agent-based models serve as a critical tool for strategizing delay mitigation, offering a balanced perspective between the granularity of analysis and the practical challenges of model implementation.

#### 4.5.2. General simulation-based models

Contrary to agent-based models that emphasize the simulation of individual agents and their interactions within a specific environment, general simulation-based models comprise an extensive array of methodologies employed to depict and analyze the behavior of intricate systems. They typically incorporate rules governing the sequence of successive flights within the same flight chain, as well as detailed parameters for flight transfers, including various assumptions and conditions (Hao et al., 2014). These

**Table 7**
Summary of the characteristics and classification of agent-based models.

| Literature | Agent definition | Data | Interaction relationship | Perspective classification |
|---|---|---|---|---|
| Fleurquin et al. (2013) | Flight | BTS data | Previous flight + Flight duration | Analysis of factors on the impact of delay propagation |
| Ciruelos et al. (2015) | Flight | CODA data | Airport | Predict the delay process by simulating the delay propagation |
| Campanelli et al. (2016) | Flight | BTS data and CODA data | Previous flight + Airport | Predict the delay process by simulating the delay propagation |
| Guleria et al. (2019) | Flight | ADS-B data | Previous flight + Flight duration + Airport | Predict the delay process by simulating the delay propagation |
| Qin et al. (2019) | Flight | VariFlight data | Previous flight + Flight duration + Airport | Analysis of factors on the impact of delay propagation |
| Wang et al. (2021) | Airport | BTS data | Previous flight + Flight duration + Airport | Predict the delay process by simulating the delay propagation |
| Cai et al. (2021b) | Flight | BTS data and ADS-B data | Flight duration | Predict the delay process by simulating the delay propagation |

models simulate the origin and spread of delays by adjusting various operational parameters such as aircraft turnaround times and weather conditions, enabling the exploration of different delay scenarios and the assessment of various management strategies to mitigate delays.

In the context of delay propagation modeling and simulation, a substantial body of research is dedicated to investigating the transmission of delays throughout the National Airspace System (NAS) and pinpointing the critical factors that trigger such propagation. Chatterji and Sridhar (2005) utilize refined meteorological indicators, specifically the Weather Impacted Traffic Index (WITI), to evaluate delay states in the NAS. By comparing these indicators with actual delay figures, they demonstrate that weather is a primary factor influencing delays. Baden et al. (2006) confirm the "ripple" effect within the NAS, attributable to the numerous flight legs each aircraft undertakes. They employ a backtracking algorithm to trace the cumulative delays back to delays in preceding flight legs, highlighting the intricate interdependencies in flight delay propagation. Barnhart et al. (2014) delve into passenger-centric delays, using a multinomial logit model combined with various data sources. Their work focuses on calculating historical passenger delays in the NAS and analyzing factors contributing to these delays. Lastly, Hao et al. (2014) dedicate their research to analyzing how delays at New York airports impact other airports within the NAS, thereby substantiating the significant role played by New York's airports in the propagation process.

Several simulation techniques, including Markov Chain and discrete-event simulation methods (Wu, 2005), mathematical decomposition (Myers et al., 2008), and parallel cellular automata (Jianli et al., 2009), have been proposed to analyze delay propagation. Specifically, Wu (2005) models the entire process of an aircraft completing its flight itinerary, including passenger and cargo/baggage processing, while considering the uncertainties in turnaround processes. This approach is used to simulate the delay propagation process and assess the stability of flight schedules. Baden et al. (2006) also focus on aircraft operational activities, employing a backtracking algorithm to trace delays back to previous flight legs under varying weather conditions. Jianli et al. (2009) explore the interconnected operational activities of multiple aircraft, establishing state space representations for different aircraft subsystems to simulate the evolution of delay propagation across subsystems. Similarly, Hao et al. (2014) concentrate on the simulation of aircraft activities in the entire NAS system, utilizing the FAA's System Wide Analysis Capability (SWAC) for rapid simulation of how delays at specific airports propagate throughout the NAS. Additionally, Myers et al. (2008) focus on airborne delays, considering detailed aircraft operational processes like vectoring, airspeed controls, directs, and amendments and their impact on airborne delay propagation. These simulation methods can be tailored to specific problems under study, with each technique offering its own set of strengths and limitations. We summarize the related studies and their findings in Table 8.

General simulation-based models are versatile tools for simulating a variety of flight delay scenarios, often backed by large-scale simulation software. While they excel in modeling delays across broad contexts due to their adaptability, they might not delve into the granular details of individual behaviors as agent-based models do. Despite this, their capacity to cover a wide spectrum of delay types makes them invaluable for overarching delay analysis and strategy formulation, albeit with a macro-level focus.

### *4.6. Queueing models*

Queueing models are a mathematical discipline that examines the behavior of queues or waiting lines. They can be applied to model flight delay propagation in air transportation networks. In this context, flights can be considered as "customers", while runways work as "servers" where flights are processed as departure and arrival flow. The goal is to understand how delays at one airport can propagate through the network and affect the performance of other airports and flights.

To be more specific, when quantifying the generated delay and its propagated effect, an iterative algorithm that operates within a 24-h cycle is commonly practical and efficient. In this widely used algorithm, a 24-h cycle is divided into multiple time windows of 5-min, 10-min, or 15-min lengths. In each time window, the newly generated delay is estimated using different queueing models for each airport in the network. After calculating the delay, its effect is propagated within the network during the specific time window,

**Table 8**
Summary of the characteristics and description of general simulation-based models.

| Literature | Simulation tools/methods | Data | Modeling perspectives | Target |
|---|---|---|---|---|
| Wu (2005) | Markov Chain algorithm + Discrete-event simulation | Airline data | Flight chain | Investigate the inherent delay and reliability of airline schedules |
| Baden et al. (2006) | Backtracking algorithm | ASQP data | Flight chain | Identify how delay propagates in NAS and it occurs in the fight legs |
| Myers et al. (2008) | Mathematical decomposition | ETMS data | Airport network | Measure and detect the airborne delay |
| Jianli et al. (2009) | Parallel cellular automata | Simulation data | Flight chain | Describe and predict of delay propagation |
| Hao et al. (2014) | NAS-wide simulation model | NOAA data + ASPM data + ASQP data | Airport network | Evaluate the impact of delay in New York airports on the rest of NAS |

Note that certain studies concentrate on NAS delay propagation rather than the simulation tools; therefore, we do not include these works in the table.

updating the information at each airport node until the algorithm processes all time windows. Consequently, the most critical tasks include selecting a suitable queueing model for delay propagation, constructing a network that integrates these queues, and applying an algorithm to ascertain whether the delay impacts other nodes in the network (Pyrgiotis et al., 2013).

***Queueing Models in Delay Quantification:*** The idea that a queueing model is applicable for modeling delay generation at each airport is inspired by the fact that most of the time, delays result from an imbalance between an airport's capacity and demand. In other words, the extra waiting time at arrival and departure runways and the turnaround time, which are the main sources of delay, can be quantified using queueing models in a queue where runways are the servers and flights are the customers.

In the process of applying queueing theory to quantify delays in each airport, the structure of the queue and the roles within it become increasingly complex and specialized. The $M/D/1$ model, in which the service time follows a uniform distribution, is a simple model applied to measure the delay propagation. The author applies this model by treating the entire airport as the server and allowing the maximum queueing length to be infinite (Hansen, 2002). However, the service speed is assumed to be constant, which does not align with reality. Daniel and Harback (2008) use the $M/M/k$ stochastic queueing model to seek the socially optimal equilibrium, where there are $k$ servers, representing the maximum capacity, to construct the congestion function. In this function, the queue evolves according to a time-variant transition matrix $T(\lambda(t), d, r, k)$, where $\lambda(t)$ is the service rate for each time-window, $d$ is the constant service rate, and $r$ is the number of runways. Furthermore, they employ congestion functions to illustrate how the probability of queue length changes in response to delay propagation. However, this paper does not clarify which part provides the service, a focus that subsequent research will need to address.

Long and Hasan (2009) treat the runways and reservoirs as consecutive servers in queues. After landing at an airport, the aircraft will be served in two stochastic queues: the arrival queue and the departure queue, either consisting of more than one server, connected by a constant turnaround delay. For the arrival queue, the servers are the arrival runway and the taxi-in queue; for the departure queue, they include the ready-to-depart reservoir, taxi-out server, and the departure runway. This model also accounts for delay propagation and potential cancellations.

Long and Hasan (2009) specify the servers in an airport as different types of runways, but group them into arrival queues and departure queues, which represents an ideal case. Considering the fact that a runway could serve either arriving aircraft or departing aircraft based on the flight flow, the Approximated Network Delay (AND) model, first conceptualized by Malone (1995), calculate the generated delay using the $M(t)/E_k(t)/1$ model in a queueing engine (QE) (Pyrgiotis et al., 2013). This study integrates the AND with the QE to estimate local delays at network nodes and to propagate delays across the system. Given the complexity of the queue, the model assumes that there is only one runway in each airport for convenience, and all aircraft follow the first-in-first-out rule, sharing the same runway within an airport. Additionally, the model approximates the expected waiting time $W(t)$ by the Little's Law:

$$W = \frac{L}{\mu} \tag{7}$$

where $L$ is the expected waiting time in the queue and $\mu$ is the service rate of the runway.

To account for the impact of taxi-out time on airport surface congestion, Simaiakis and Balakrishnan (2016) introduce an approach that combines taxi-out time distribution estimations with the $D(t)/E_k(t)/1$ queueing model, specifically aiming at the aircraft departure process. This method enables accurate prediction of taxi-out time, departure volume, and delay magnitude for the two runway configurations at the targeted airport. Subsequent studies have continued to use combinations of previous queueing models and approximated Little's Law to measure the delay when applying the queueing theory. Baspinar et al. (2016) modify the AND model for European airports and use the $M(t)/E_k(t)/1$ model to capture the effect of local disturbances. Wu et al. (2018a) construct the Airport-Sector Network Delays (ASND) model for calculating the generated delay based on the AND model and the Link Transmission Model (LTM), which considers flight delays caused by the airspace constraints in each airport. Subsequently, they employ a delay propagation algorithm that updates flight itineraries and demand rates at each airport. This update is based on local delays computed by the queuing engine, as well as on flow control delays. Zhao et al. (2022) choose to use $M/G/1$ model, a combination of $M/E/1$ system, to capture the propagation delay in a multiple-airport system (MAS). However, research following

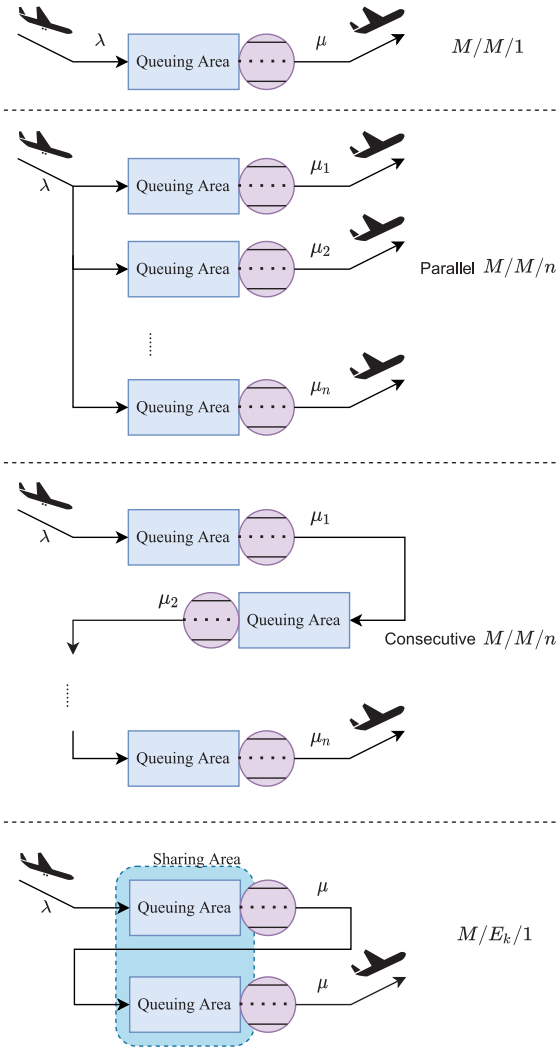



**Fig. 10.** An illustration of representative queueing models.

the AND model has done little to explore new queueing models suitable for measuring generated delay in each node of a system; instead, most innovations have focused on the structure of the entire network and how to design the iteration algorithm. The illustration of representative queueing models can be found in Fig. 10.

***Network Connecting the Queues***: In a network system, nodes and edges are fundamental components for representing the structure. It is natural to construct a network structure to describe delay propagation within an airport system, since the aircraft, the main source of delay propagation, connects airports through flights, crews, and passengers. In this case, a well-structured network is more applicable to queueing theory. A simple network includes nodes for each airport and edges for flights. For example, the LMINET2 model propagates delays between nodes through edges that transfer the number of flights in time windows (Long and Hasan, 2009). However, a more efficient way to define edges appears in the AND model, in which the Delay Propagation Algorithm (DPA) defines edges as channels that transfer the arrival rate between two neighboring nodes and two consecutive time windows (Baspinar et al., 2016).

In the process of delay modeling, it is of paramount importance to consider en-route delays, as these periods are often associated with airport arrival and departure waypoints. Furthermore, the capacity constraints of air routes contribute to the persistent occurrence of flight delays. In this regard, Lin et al. (2021) develop a novel Multi-layer Air Traffic Network Delay (MATND) model, which considers both the airport and en-route congestion points as single servers and proposes a dynamic queueing model on the constructed network.

**Table 9**
Model features for the literature reviewed in queueing models.

| Literature | Queueing model | Servers' type | | | Departure & Arrival runways | |
|---|---|---|---|---|---|---|
| | | Single | Multiple parallel | Multiple consecutive | Shared | Separated |
| Hansen (2002) | $M/D/1$ | ✓ | | | | ✓ |
| Daniel and Harback (2008) | $M/M/k$ | | ✓ | | | ✓ |
| Long and Hasan (2009) | $M/M/k$ | | | ✓ | | ✓ |
| Pyrgiotis et al. (2013) | $M/E/1$ | ✓ | | | | ✓ |
| Simaiakis and Balakrishnan (2016) | $D/E/1$ | ✓ | | | ✓ | |
| Baspinar et al. (2016) | $M/E/1$ | ✓ | | | ✓ | |
| Wu et al. (2018a) | $M/E/1$ | ✓ | | | ✓ | |
| Lin et al. (2021) | $M/E/1$ | ✓ | | | ✓ | |
| Zhao et al. (2022) | $M/G/1$ | ✓ | | | ✓ | |

With the introduction of MAS, Zhao et al. (2022) expand the nodes from solely airports to a system of arrival waypoints, departure waypoints and airports. This definition of nodes is more suitable for the reality where airports in the same area share the same waypoint. These are the main popular network models related to queueing models, but the networks themselves can have more diverse structures, which can be seen in the network representation model Section 4.7 when other theories are applied. Literature and summaries pertaining to the utilization of queueing models for modeling flight delay propagation are exhibited in Table 9.

Queueing models provide a structured framework to systematically explore how factors like runway capacity and aircraft queuing affect delays, offering a clear and theoretical skeleton for understanding bottlenecks and delay propagation at specific points, such as runways. While adept at addressing airport and aircraft-related delays, these models inherently focus on operational chokepoints, which might not encompass all complexities of air traffic systems, such as external influences like weather or seasonal variations. Despite this, queueing models' theoretical approach makes them particularly suitable for analyzing and managing delays at targeted locations within the airport system, providing valuable insights for optimizing flow and reducing congestion.

### *4.7. Network representation models*

Network representation models can be generally grouped into Bayesian networks, complex networks, graph neural networks and alternative spatio-temporal models based on the perspectives of network modeling and the characteristics of these models.

#### *4.7.1. Bayesian networks*

Bayesian networks (BN) are probabilistic models that encode information about random variables and their interrelationships into the nodes and arcs through a directed acyclic graph. They combine the benefits of traditional probabilistic methods, interpretability, visualization, and computational efficiency (Rodríguez-Sanz et al., 2019). Bayesian networks are ideal tools for systematically modeling uncertainties and investigating interrelationships among multiple factors in complex systems. Therefore, they are broadly applied in the field of delay propagation modeling. The main aims of these researches include: (1) Investigating the delay propagation among airports (Xu et al., 2005, 2007); (2) Exploring the sequential flight delay (Cao et al., 2008); (3) Estimating delays in a busy hub-airport and discuss the effect of propagation within and from this airport (Liu et al., 2008); (4) Investigating the impact of different factors on delays (Liu and Ma, 2008b); (5) Comparing different models and improve the learning algorithms (Liu and Ma, 2008a; Liu and Yang, 2009; Liu and Ma, 2009; Cao and Fang, 2012; Liu and Wu, 2013). Table 10 surveys recent studies on delay propagation problems under the BN framework.

The construction of a Bayesian network model, often executed using Netica Software, encompasses three primary steps:

(1) Variable Creation: This involves identifying relevant variables such as flight departure delay, inbound flight delay, turnaround process delay, pilot connection delay, cabin crew connection delay, and passenger connection delay, among others (Wu and Law, 2019).
(2) Structure Learning: This step may be informed by expert judgment or derived from various algorithms.
(3) Parameter Learning: Typically implemented using the Expectation-Maximization (EM) algorithm.

For in-depth insights into structure learning and parameter learning as discussed in existing literature, readers are referred to Table 10.

In most of the early literature, the BN structure is established based on expert judgment and available data. Regression models can aid researchers in making informed decisions (Laskey et al., 2012). However, expert knowledge may not always be reliable, necessitating structure learning in BN construction. Liu and Ma (2008a) utilize a score-based greedy search algorithm published in 1992 (Cooper and Herskovits, 1992), called K2, achieving higher accuracy in their model. Nevertheless, this traditional algorithm exhibits low learning efficiency and high sensitivity to the order of input attributes. Subsequent research has focused on enhancing the K2 algorithm. In a later study by Liu and Yang (2009), a Target-fixed Stochastic-ordered K2 (TSK2) Algorithm is proposed and demonstrated to be well-suited for modeling flight delay propagation. This approach offers increased flexibility and efficiency. Alternative algorithms have also been employed for structure learning in BN. Liu and Wu (2013) find that despite improvements in algorithms, certain specific flight records could not be accommodated within the model. Consequently, they proposed a new

**Table 10**
Recent studies on delay propagation problems under BN framework.

| Literature | Modeling target | Network construction | Parameter estimation | Other methods combined |
|---|---|---|---|---|
| Xu et al. (2005) | Propagation of delay between airports | The initial BN model structure was developed using expert judgment. A statistical significance test was then conducted on pairs of nodes connected by an arc in the expert-elicited BN. | The initial exploratory parameter estimation was performed using Netica's parameter learning capability. Then regression analysis was used to estimate the parameters. | / |
| Cao et al. (2008) | The sequence flight delay and delay propagation of the airline company | The structure of the model is developed based on experts' experience. | Discretize the data and use Netica's parameter learning capability | / |
| Liu et al. (2008) | Arrival delay and the propagation delay of a busy hub-airport | The model structure is developed based on experts' experience and regression analysis. | Discretize the data and use Netica's parameter learning capability | / |
| Liu and Ma (2008b) | Departure delay of a flight | The structure of all the models is developed based on available data and experts' experience. The causal relationships between the variables in the BN model are from past literature. | Discretize the data and use Netica's parameter learning capability. For incomplete data in records, Expectation-Maximization (EM) arithmetic is used to learn BN parameters. | / |
| Liu and Ma (2008a) | Flight delay in a busy hub-airport | Use K2 algorithm to learn the topology and then modify the flight chain model structure artificially | EM algorithm is used. | / |
| Liu and Yang (2009) | Initial delay of a flight | Use Target-fixed Stochastic-ordered K2 Algorithm to model | Discretize the data and use Netica's parameter learning capability | / |
| Liu and Ma (2009) | Flight delay in a busy hub-airport | Use K2 algorithm (a score-based greedy search algorithm) in this part | EM algorithm is used. | / |
| Laskey et al. (2012) | The relationships among different components of aircraft delay and the causal factors that affect delays | The model structure is developed based on experts' experience and regression analysis. | Estimate initial local distributions for the given node by discretizing the regression model; use Dirichlet-multinomial learning from the training data to update the distributions of all nodes. | / |
| Cao and Fang (2012) | Flight delay in a busy hub-airport | High Score Prior Genetic-Simulated Annealing Approach to Bayesian Network Structure Learning | EM algorithm is used. | / |
| Liu and Wu (2013) | Build a model that can accommodate unusual flights | Negotiating Method with Competition and Redundancy | EM algorithm is used. | / |
| Kaya and Inalhan (2014) | Predict delay states of a route more accurately | The model structure is designed by professional softwares and expert knowledge. | Gradient descent is used. | Dynamic Bayesian Belief Network |
| Wu and Wu (2018) | Multi-flight delay propagation and delay interdependencies | The model structure is developed based on experts' experience. | Calculate the prior probability distributions of each variable and their joint probability distributions for each flight in the model based on historical data | Delay propagation tree |
| Wu and Law (2019) | A non-IID based Bayesian Network model in a delay tree framework to model delay propagation by considering several factors | Develop a delay propagation tree model with Bayesian Network (DPT-BN) by combining the concept of DPT and BN in a single model | Based on historical data, Netica's parameter learning capability is used. | Multiple resource connections and delay propagation tree |
| Li and Jing (2021) | The complex phenomenon of delay propagation within a large network | Apply a scoring function and a search strategy to traverse the search space of possible models | Nodes are parameterized by the prior probability and the conditional relationship. A data-driven method is used. | Complex networks |

algorithm called the Negotiating Method with Competition and Redundancy (NMCR) for modeling arrival flight delays at busy hub airports. Although the resulting BN exhibits a more complex topology, its operational speed is demonstrated to be acceptable. Cao

and Fang (2012) develop an algorithm called the high score prior genetic-simulated annealing (HSPGSA) approach for the network structure learning. This method combines the principles of genetic algorithms (GA) and simulated annealing (SA) algorithms, offering the advantages of high accuracy and rapid operational speed compared to the standalone GA and SA algorithms.

For parameter learning, Netica assumes multinomial random variables with Dirichlet conjugate prior distributions, which are appropriate for categorical random variables with unrelated parameters. To circumvent the need for discretizing continuous variables, the EM arithmetic is applied in parameter learning (Liu and Ma, 2008b,a). However, EM may suffer from large computational scales and slow convergence. To address this issue, Liu and Ma (2009) separate the estimation model for arrival delay into two unattached models. One model contains time-related information, while the other incorporates flight-related information.

In recent years, studies have focused on improving Bayesian networks for flight delay modeling and prediction. For example, Kaya and Inalhan (2014) propose a Dynamic Bayesian Belief Network (DBBN) that considers temporal, operational, spatial, traffic, and hidden variables to enhance delay prediction performance at specific times prior to the scheduled departure. Some researchers have combined BN with delay propagation trees (DPT). While DPT models often adopt independent and identically distributed (IID) assumptions, Wu and Wu (2018) explore the non-IID nature of delay time distributions based on real airline data. They develop an enhanced Delay Propagation Tree model with Bayesian Networks (DPT-BN) that combines the advantages of DPT and BN, resulting in better handling of stochastic variables and non-IID effects of delay propagation, as well as improved network visualization. In another study, Wu and Law (2019) propose a DPT-BN to explore propagation effects, diagnose current airline operations, and seek improvement measures for flights' on-time performance.

In addition to the combination of BN and DPT, researchers have also integrated BN with complex network theory. Li and Jing (2021) utilize complex network theory to conduct a macroscopic examination of the topological characteristics of propagation networks after capturing the conditional dependence of flight delays through Bayesian networks. This approach allows for the exploration of issues like airport and route importance, the shortest path for delay propagation, and more. Researchers have also employed BN to investigate the impact of delays at upstream and downstream airports, simulating the propagation of uncertainties when delays occur at different locations. This method offers a more comprehensive understanding of complex system-level delay propagation compared to pairwise analysis.

Indeed, BN has been extensively utilized to elucidate delay propagation patterns and quantify the influence of variables in the aviation industry. By capturing the interdependencies between factors that affect performance and delay, researchers can gain a better understanding of how these variables interact and influence one another. Rodríguez-Sanz et al. (2018) develop BN to analyze aircraft flow within the airport operational framework, allowing them to uncover the relationships between factors that impact performance and delay. In addition to comprehending the underlying patterns of delay propagation, BN can also be used to visualize these relationships. To be specific, researchers can first construct a Bayesian network to analyze the delay parameters and select delay factors for visualization. Then, a series of visualization methods can be employed to present the propagation of flight delays and the analysis is available (Chen et al., 2020).

Bayesian networks employ probabilistic modeling to reason about delays, offering a structured approach to navigate the uncertainties inherent in air traffic data. This method excels in delineating the probabilistic relationships and underlying causes of delays, allowing for a clear exploration of the impacts between various variables within the system. However, constructing and interpreting these networks requires a significant level of expertise and industry-specific knowledge, marking a potential barrier to their widespread application. Despite this, Bayesian networks' ability to systematically uncover and analyze the intricate web of causal relationships makes them particularly suited for studies focused on understanding the probabilistic nature of flight delays.

#### *4.7.2. Complex networks*

The air traffic network embodies a highly dynamic and intricate system. Owing to its low complexity, robust interpretability, and adaptable modeling, complex network theory has been integrated into this field, facilitating a more profound exploration of the network structure and dynamics (Zhixing et al., 2021). The complex networks modeling approach for air traffic is structured predominantly in two steps:

(1) Network Construction: This step involves abstracting and simplifying the air traffic network by representing airports as nodes and encoding traffic information into the edges. The construction of these networks is based on real flight data, drawing from diverse regions like the US, China, and Europe.
(2) Network Analysis: Analysis of these constructed networks hinges on metrics derived from network theory. While some studies utilize fundamental metrics of complex networks, as detailed in Table 11, others introduce novel evaluation methods. By applying network analysis techniques, researchers are able to identify key characteristics of the networks and uncover causal relationships in flight delay patterns.

***Network Characteristics Analysis***: In the realm of network structure analysis, the conventional approach encompasses the "flight network" configuration, which abstracts the airport network into a complex system, wherein airports are denoted as nodes and direct flights as edges. Fleurquin et al. (2014b) employ this model, incorporating delay information by adjusting node sizes. Conversely, Wang et al. (2020b) put forth an alternative weighted airport delay network, wherein airports are still represented as nodes; however, links are established in the presence of direct flights between two airports, with the aggregate of delays between them serving as the edge weight. By juxtaposing the disparities in the betweenness centrality of the two networks, Wang et al. (2020b) demonstrate that their method enables a more incisive examination of the attributes inherent to complex systems.

The foundational structural properties, encompassing aspects such as degree distribution and centrality, are scrutinized within these scholarly contributions. Fleurquin et al. (2014b) showcase a heterogeneous network, characterized by a limited number of

**Table 11**
Topological metrics for complex network analysis.

| Metrics | Formula | Interpretation | Literature |
|---|---|---|---|
| Degree | In-degree of airport i: $d_i^{in} = \sum_{j=1}^{N} a_{ji}$<br>Out-degree of airport i: $d_i^{out} = \sum_{j=1}^{N} a_{ij}$ | Inflow delays at airport i<br>Outflow delays at airport i | Du et al. (2018), Xiao et al. (2020), Wang et al. (2020b), Fleurquin et al. (2014a,b) and Zanin et al. (2017) |
| Reciprocity parameter | $R = \frac{\sum_{i\neq j}^{N}(a_{ij}-\bar{a})(a_{ji}-\bar{a})}{\sum_{i\neq j}^{N}(a_{ij}-\bar{a})^2}$, where $\bar{a} = \frac{\sum_{i\neq j}^{N} a_{ij}}{N(N-1)}$ | The bidirectional properties of delay propagation links between airport pairs | Du et al. (2018) |
| Clustering coefficient | $C^D = \frac{1}{n}\sum_{i=1}^{n} \frac{\left(\frac{1}{2}\right)\sum_j \sum_h (a_{ij}+a_{ji})(a_{ih}+a_{hi})(a_{jh}+a_{hj})}{[d_i^{dot}(d_i^{dot}-1)-2d_i^{\leftrightarrow}]}$<br>where $d_i^{dot} = \sum_{j\neq i} a_{ji} + \sum_{j\neq i} a_{ij}$,<br>$d_i^{\leftrightarrow} = \sum_{j\neq i} a_{ij}a_{ji}$ | The inherent cluster tendency of airports | Du et al. (2018), Xiao et al. (2020) and Wang et al. (2020b) |
| Largest connected cluster | The airport i is considered as a member of the cluster when $d_i^{\text{out}} \geqslant k^{\text{out}}$ where the threshold $k^{\text{out}}$ is a constant | The extent of delay propagation | Du et al. (2018) |
| Community(Modularity) | First divide the delay propagation between sub-airports into several sub-areas, and each sub-area has dense delay propagation links inside the airport, then<br>$Q_d = \frac{1}{M}\sum_{ij}\left(A_{ij} - \frac{d_i^{\text{out}} d_j^{\text{in}}}{M}\right)\delta(c_i, c_j)$ | It is used to evaluate whether the delay propagation among airports can be divided into several sub-regions. | Du et al. (2018) and Xiao et al. (2020) |
| Network motifs | Z-score - the significance level of motifs:<br>$Z(G') = \frac{F_G(G')-\mu_R(G')}{\sigma_R(G')}$, where $F_G(G')$ denotes the frequency of $G'$ in network $G$ and $\mu_R(G')$ and $\sigma_R(G')$ represent the mean and standard deviation of the frequency in a set of N random networks. | The local relationship pattern among any three airports | Du et al. (2018), Xiao et al. (2020) and Zanin et al. (2017) |
| Link density | $L = \frac{\sum_{i\neq j}^{N} a_{ij}}{N^2} = \frac{E}{N^2}$ | The proportion of active links to the total number of potential links | Xiao et al. (2020) and Zanin et al. (2017) |
| Assortativity coefficient | $S = \frac{1}{E}\sum_{j>i}\frac{1}{2}(d_i + d_j)a_{ij}$, where $d_i$ represents the degree of airport $i$ including the in-degree and out-degree. | It is used to examine whether vertices with similar degrees tend to be connected. | Xiao et al. (2020), Wang et al. (2020b) and Zanin et al. (2017) |
| Efficiency | $E = \frac{1}{N(N-1)}\sum_{i\neq j}\frac{1}{e_{ij}}$, where $e_{ij}$ is the distance between airport i and airport j. | The number of intermediate nodes which delays pass through to reach the target node | Xiao et al. (2020) and Zanin et al. (2017) |
| Rich-club coefficient | $\phi(k) = \frac{2E_{>k}}{N_{>k}(N_{>k}-1)}$, where $E_{>k}$ is the number of edges between the nodes of degree greater than or equal to $k$, and $N_{>k}$ is the number of nodes with degree greater than or equal to $k$. | The extent to which well-connected nodes also connect to each other | Wang et al. (2020b) |
| Betweenness centrality | $g(v) = \sum_{s\neq v\neq t}\frac{\sigma_{st}(v)}{\sigma_{st}}$, where $v$ denotes the node $v$, $\sigma_{st}$ is the total number of shortest paths from node $s$ to node $t$ and $\sigma_{st}(v)$ is the number of those paths that pass through $v$ | The number of these shortest paths that pass through the node $v$ | Wang et al. (2020b) |
| Transitivity | $T = \frac{3N_\Delta}{N_3}$, where $N_\Delta = \sum_{k>i>j} a_{i,j}a_{i,k}a_{j,k}$,<br>$N_3 = \sum_{k>i>j}(a_{i,j}a_{i,k} + a_{j,i}a_{j,k} + a_{k,i}a_{k,j})$ | The presence of triangles in the network which indicates that groups of three airports are strongly connected | Zanin et al. (2017) |
| Diameter | The greatest distance between any pair of nodes | The number of flights needed for delays to spread across the whole system | Zanin et al. (2017) |

$A_{ij=}(a_{ij})_{N*N}$ is the adjacency matrix of the network. $a_{ij} = 1$ when there exists an edge from node $i$ to node $j$, otherwise $a_{ij} = 0$.
$N$ represents the number of nodes in the network; $E$ represents the number of links in the network.

airports responsible for the majority of traffic. Wang et al. (2020b), on the other hand, present quintessential airport network features, including small-world properties, scale-free attributes, and disassortativity characteristics. Additionally, the rich-club coefficient and betweenness centrality are computed to discern the core airports within the delay network, further elucidating the intricate dynamics that govern air traffic networks and delay propagation.

In the context of delay propagation characteristics, numerous aspects such as flight chains, robustness, temporal properties, and system saturation are taken into account. Fleurquin et al. (2014b) incorporate flight chain analysis to delineate the progression

of reactionary delays within their network. Another study by Fleurquin et al. (2014a), proposes a metric for assessing network robustness and delay impact, scrutinizing the repercussions of disturbances on the system from both airport and flight perspectives, while comparing variances in disturbances at disparate times of day. Several works, such as those by Wang et al. (2020b) and Cai et al. (2021b), delve into the temporal features of the system. Wang et al. (2020b) employ a technique dubbed "Multifractal Detrended Fluctuation Analysis" to examine flight delay series, exploring the fractal properties of delay propagation and the system's capacity to manage unanticipated delays. Cai et al. (2021b) construct a spatio-temporal delay network to further probe the system's dynamics. They propose three indicators – magnitude, severity, and speed – to quantify the number of delays, delay duration, and the pace of delay propagation, respectively. Regarding system saturation, which pertains to the air traffic system's insufficient capacity to maintain timely aircraft flows, Rodríguez-Sanz et al. (2018) put forward various indices and performance indicators to quantify delay patterns at each stage of an aircraft's journey (arrival, taxi-in, turnaround, taxi-out) and the degree of system saturation. Contrasting with the analysis of network structure characteristics, these studies progressively incorporate the dynamic nature of the system.

***Flight Delay Causal Relations***: While numerous studies elucidate the attributes of delay networks through the abstraction of air traffic networks into complex systems, a substantial portion of these investigations construe the network as static, neglecting the interdependence of delay time series. In order to more effectively discern the causality relations inherent to flight delay time series, certain research endeavors to construct a delay causality network, examining the dynamic progression of delay propagation between airports. In these models, the input consists of airport delay time series, while the output comprises the delay causal network. Nodes within the network represent airports, and links between node pairs correspond to causality relations, identified via the application of statistical significance tests. Given the complexity of air traffic networks and the multitude of airports involved, complex network metrics are also employed in the analysis of causal delay networks, thereby unveiling the network's characteristics and delineating the propagation scenario.

The primary distinctions between various studies lie in the significance test employed. The Granger causality test is the most prevalent method, as it can discern true causalities while excluding false correlations. This approach has been adopted by Zanin et al. (2017), Du et al. (2018) and Pastorino and Zanin (2021) in their investigations of aviation systems. However, Granger causality is not without its limitations:

(i) The Granger causality test is tailored to identify linear relations, while flight delays tend to propagate in a non-linear fashion.
(ii) Owing to the utilization of averaged and smoothed data, Granger causality may overlook substantial anomalous delays, thereby diminishing its statistical significance in delay propagation (Zanin et al., 2017).

Additionally, Mazzarisi et al. (2020) introduce an expanded application of the Granger causality test, termed "Granger causality in the tail". This method endeavors to investigate the relationships between sizable delays that arise under extreme events. Moreover, the outcomes derived from this novel approach are compared with those obtained through the conventional Granger causality test, offering a more nuanced understanding of delay propagation under various circumstances.

Owing to the linearity of the Granger causality test and the high-dimensional nature of delay time series, Xiao et al. (2020) propose the low-dimensional approximation of high-dimensional conditional mutual information for transfer entropy as a means to construct a delay causality network. By comparing the results of the Granger causality test with those of the transfer entropy method, it becomes evident that this refined algorithm facilitates the construction of more realistic delay causal networks and accurately uncovers the true dynamic processes of delay propagation between airports. This issue can also be partially addressed by employing information-based nonlinear Granger causality; however, biases may arise when evaluating high-dimensional conditional mutual information (CMI). In 2022, Jia et al. (2022) propose the nonlinear Granger causality method, which utilizes a low-dimensional approximation CMI (LA-CMI) to calculate a more precise estimation of nonlinear multivariate causality, thereby circumventing the curse of dimensionality. The validity of this approach is substantiated through the analysis of simulated multivariate time series in their study.

Following the establishment of the delay causal network, a characteristic analysis can be conducted to investigate delay propagation. The analysis properties can be classified into global, local, and layered properties. Predominantly, global property analysis is employed to account for system-level network characteristics by computing topological metrics as shown in Table 11 (Du et al., 2018; Xiao et al., 2020; Zanin et al., 2017; Pastorino and Zanin, 2021). Local property analysis leverages the motifs in Table 11 to examine variations in delay propagation characteristics for airports of different sizes (Du et al., 2018; Xiao et al., 2020). Layered property analysis, as performed by Zanin et al. (2017) and Du et al. (2018), involves dividing the entire network into layers based on airlines and subsequently conducting property analysis. Each layer of the network represents an airline, with all layers collectively forming the entire network; certain edges that appear in individual layers may not be present in the overall network. This approach aims to uncover the contributions of airlines to delay propagation. In addition, Guo et al. (2022) establish a two-stage analytical framework to detect delay propagation. Initially, a delay causality devoid of the strong and independent causality (SIC) assumption is obtained via a convergent cross-mapping method to represent the delay propagation effect. In the second stage, a delay causality network based on the non-SIC assumption (DCG-non-SIC) is constructed. A complex network analysis of DCG-non-SIC is then presented to reveal delay propagation patterns, yielding some counterintuitive findings. For example, by calculating the eigenvector centrality, the authors find that the importance of airports in delay propagation is not exactly in line with the hub class or the airport size.

Complex networks in modeling flight delay propagation leverage network theory to examine the interconnectedness of the air traffic system, providing a macro-level analysis of how delays spread across the network. This approach is particularly adept at identifying systemic vulnerabilities and assessing network resilience, offering insights into the structural dynamics that facilitate

delay propagation. However, while excelling in broad, network-wide analysis, complex networks may not adequately account for the nuances of individual factors or external influences on flight delays. Their high-level perspective is instrumental in understanding the overarching characteristics of constructed delay networks but might miss the detailed impact of specific operational or environmental variables on delays. Despite this limitation, complex networks are invaluable for strategic, system-wide delay analysis, enabling stakeholders to identify critical points for intervention and enhance overall network efficiency.

#### 4.7.3. Graph neural networks

Graph neural networks (GNNs) constitute a category of deep learning paradigms, specifically devised for manipulating graph-structured data. Graph neural networks have enjoyed extensive application in domains such as transportation, biomedicine, and social networks, owing to their potent representational capabilities for non-Euclidean data and huge potential in information aggregation. It is noted that, graph neural networks have been substantiated as eminently suitable for traffic flow forecasting tasks, given their capacity to capture and model the dynamism and uncertainty inherent in spatio-temporal traffic flow (Jiang and Luo, 2022). Besides, as indicated in Li et al. (2021a), graph signal processing can also be harnessed to scrutinize flight delay propagation. For instance, airport delays can be viewed as node signals in a graph, allowing for the identification and quantification of various spatio-temporal patterns using graph spectral analysis. Fig. 11, adapted from (Bui et al., 2022), presents the general framework for airport network delay propagation prediction utilizing spatio-temporal graph neural networks (ST-GNN), specifically tailored for flight delay contexts. In this regard, flight delay propagation prediction tasks can be summarized in the following form:

$$\left[\boldsymbol{X}^{(t-T'+1)}, \ldots, \boldsymbol{X}^{(t)}; \mathcal{G}\right] \xrightarrow{f(\cdot)} \left[\boldsymbol{X}^{(t+1)}, \ldots, \boldsymbol{X}^{(t+T)}\right] \tag{8}$$

where $\boldsymbol{X}^{(t)} \in R^{N \times d}$ denotes the flight delay states matrix of $N$ airports at time step $t$, $\mathcal{G}$ is the established graph structure, $T$ denotes the prediction steps, and $f(\cdot)$ is the corresponding ST-GNN models for flight delay propagation prediction. The use of GNNs in modeling airport network delay propagation hinges on constructing an appropriate graph learning model that effectively captures the complex spatial and temporal dependencies among nodes. By feeding historical delay states of the airport network into the model, GNNs predict future network delay states, thereby representing the dynamic evolution of network delays through learned patterns (Bao et al., 2021).

At the time of composing this paper, there exist six publicly accessible papers that address the modeling of network delay propagation through the application of graph neural networks, as listed in Table 12. Zeng et al. (2021) develop the deep graph-embedded LSTM (DGLSTM) model, which employs a diffusion convolution kernel to capture the delay propagation characteristics, while using LSTM to model temporal dependencies. Subsequently, Bao et al. (2021) introduce an innovative graph-to-sequence learning architecture termed AG2S-Net, incorporating an attention mechanism and centering on the formulation of an adaptive adjacency matrix. In response to the time-varying and periodic information within airport networks, Cai et al. (2021a) propose the Multiscale Spatial-Temporal Adaptive Graph Convolutional Neural Network (MSTAGCN), based on the well-designed adaptive graph convolutional block. Sun et al. (2022) present the dynamic spatial–temporal graph attention neural network (DST-GAT), which employs the graph attention network (GAT) to capture the dynamic adjacency matrices. A recent study by Wu et al. (2022) suggests a space–time separable multi-graph convolutional network framework (STPN). In this investigation, temporal and spatial dependencies within the airport delay network are extracted independently. The experimental results demonstrate state-of-the-art performance across multiple datasets and multi-step predictions. Moreover, spatio-temporal graph neural networks can be employed to model flights, thereby more accurately characterizing flight attributes and facilitating downstream forecasting and classification endeavors (Guo et al., 2020).

Graph neural networks apply deep learning techniques to the study of flight delay propagation within complex network structures, offering a cutting-edge approach to deciphering the intricate behaviors of delays as they traverse the air traffic system. These models are particularly potent in analyzing large-scale network interactions, harnessing the depth and breadth of deep learning to predict delay patterns across vast networks with high accuracy. However, the sophistication of GNNs comes with the cost of significant computational resources, necessitating powerful hardware and substantial data processing capabilities. Despite this, graph neural networks stand out for their unparalleled ability to model the dynamics of flight delay propagation on a large scale, making them ideally suited for comprehensive analyses of air traffic networks where predictive accuracy is paramount.

#### 4.7.4. Alternative spatio-temporal methods

Alternative spatio-temporal modeling approaches encompass those models which, beyond the aforementioned methods, utilize networked representations to capture flight delay propagation, as illustrated in Table 13. While these methods vary in their specific network representations and analytical approaches, they share a common goal of leveraging spatio-temporal data to more accurately model and predict the propagation of flight delays. Each technique employs unique mechanisms to capture the complex dynamics of delay propagation, considering both spatial interconnections and temporal sequences in the aviation network.

Abdelghany et al. (2004) propose a flight delay projection model based on the constructed directed acyclic graph, with scheduled events serving as nodes and the various requisite activities for these events functioning as arcs. This representation allows for a structured and comprehensive understanding of the interconnected components within the flight schedule. AhmadBeygi et al. (2008) introduce a concept named Delay Propagation Tree (DPT), a framework designed to quantify and illustrate the potential extent of delay propagation throughout the network, considering pre-planned schedules and connections for cockpit crew and aircraft. In recent studies, DPT has been further enhanced and combined with Bayesian networks to analyze probabilistic delay propagation (Wu and Wu, 2018; Wu and Law, 2019).

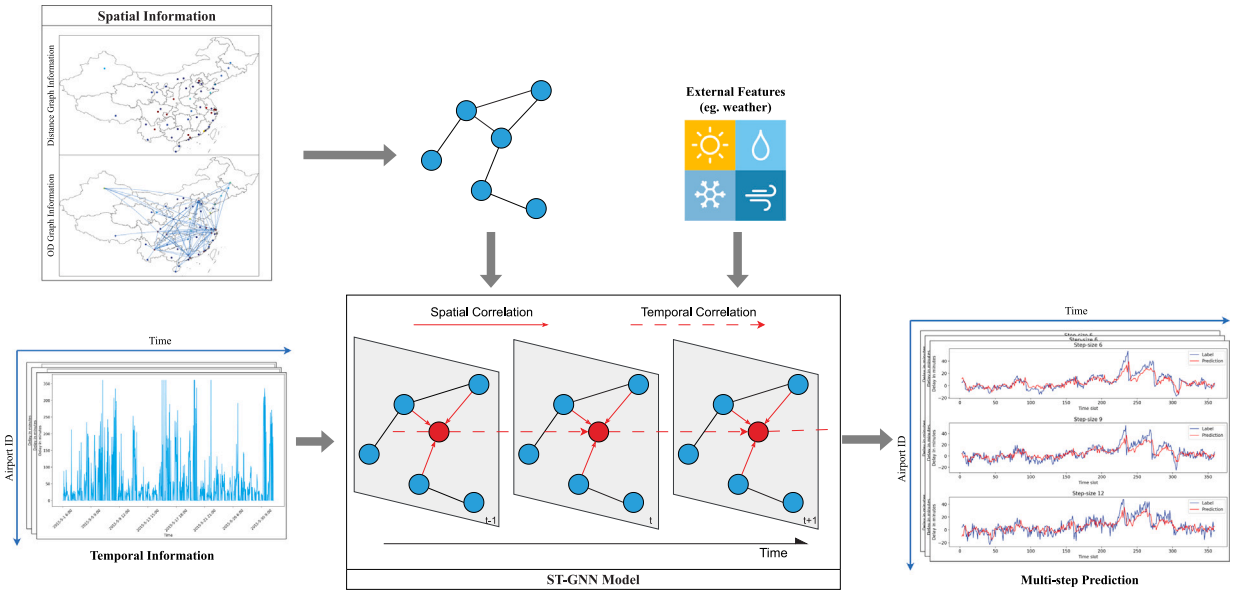



**Fig. 11.** A general pipeline of ST-GNN models for airport delay propagation prediction.

**Table 12**
Summary of the characteristics and classification of graph neural network models.

| Literature | Model | Spatial topology construction | Spatial dependency | Temporal dependency | External features |
|---|---|---|---|---|---|
| Guo et al. (2020) | SGDAN | Heterogeneous graph | Graph embedding + Attention | Attention | Weather + Flying distance |
| Zeng et al. (2021) | DGLSTM | Distance graph + O–D graph | GCN | LSTM | / |
| Bao et al. (2021) | AG2S-Net | O–D graph | GCN | Bi-LSTM + Attention | Weather |
| Cai et al. (2021a) | MSTAGCN | Adaptive graph | GCN | R-GCN | Emergencies effects |
| Sun et al. (2022) | DST-GAT | Distance graph + O–D graph | GAT | LSTM | / |
| Wu et al. (2022) | STPN | Distance graph + O–D graph + D-O graph | GCN | Position embedding + Attention | Weather |

Timed Petri Nets (TPNs) are also an effective tool for analyzing flight delay propagation, which provides a structured and formalized framework capable of capturing the complex relationships between various events and processes in aviation operations. Ding et al. (2009) pioneer the application of TPNs for elucidating the trajectory and principles governing delay propagation between two adjacent flights. An advanced variant, Time Interval Petri Nets (TIPN) proposed by Zhang and Zhu (2014), incorporates interval constraints allocated to both places and transitions, thus facilitating the modeling of turnaround time and enhancing the overall representation of flight operations. Acknowledging the detrimental consequences of uncertainty and delay probability distributions, Shao and Xu (2020) employ Colored-timed Petri Nets for modeling and examining delay propagation under various uncertain conditions in airport networks. This approach contributes valuable insights into the behavior of delay propagation in the face of diverse uncertainties, enhancing the understanding of complex airline operational dynamics.

Some advanced deep learning techniques have also been employed to uncover the complex dynamic characteristics of delay propagation in aviation networks. Ai et al. (2019) propose a convolutional LSTM network (Conv-LSTM) for airport network delay prediction, where the convolutional neural networks (CNNs) and LSTM are developed to model the spatial and temporal dependencies, respectively. To encapsulate the dynamic spatial dependencies, Zhang et al. (2021a) develop a deep learning framework, ARR-LSTM, where a PageRank module is tailored to extract dynamic adjacency matrix. Another feasible approach is to construct airport situational awareness maps using trajectory data, and then employ CNNs to derive relevant features for flight delay prediction (Shao et al., 2022). Moreover, it is verified that tensor factorization is an effective tool for extracting important features and patterns from high-dimensional delay data (Zhang et al., 2021b).

Alternative spatio-temporal methods intricately consider the spatial and temporal dimensions of delays, offering a comprehensive framework for analyzing how delays spread across both time and space. These methods excel in their ability to handle complex datasets, integrating diverse data types to map out the nuanced dynamics of delay propagation. However, the diversity of approaches under this category means that selecting the most appropriate technique often requires expert knowledge and careful consideration

**Table 13**
Summary of the characteristics and description of alternative spatio-temporal methods.

| Literature | Network construction | Modeling perspective | Data | Description |
|---|---|---|---|---|
| Abdelghany et al. (2004) | Directed acyclic graph | Flight chain | Airline data | Use shortest path algorithm to ascertain the earliest possible time for scheduled events |
| AhmadBeygi et al. (2008) | Delay Propagation Tree | Flight chain | Airline data | Utilize the "network probing" technique to depict how root delay propagates through the tree network |
| Ding et al. (2009) | Timed Petri Nets | Flight chain | Simulation data | Define the parameters in TPNs with the connection of various delay conditions |
| Zhang and Zhu (2014) | Time Interval Petri Nets | Flight chain | Simulation data | Combine with linear logic reasoning to simplify the mapping rules |
| Shao and Xu (2020) | Colored-timed Petri Nets | Flight chain | Airline data | Unveil the mechanisms of uncertain factors on the flight delay propagation |
| Ai et al. (2019) | Conv-LSTM | Airport network | Variflight data | Employ CNN and LSTM to capture spatial and temporal dependencies |
| Zhang et al. (2021b) | Tensor factorization | Airport network | CAAC data | Combine Tucker decomposition and latent class analysis to characterize the principal patterns |
| Zhang et al. (2021a) | APR-LSTM | Airport network | Variflight data | Apply PageRank and LSTM to capture spatial and temporal dependencies |
| Shao et al. (2022) | Airport situation awareness map | Airport network | BTS data | Employ CNN to extract the features in the constructed map |

of each method's characteristics. Despite this, alternative spatio-temporal methods are invaluable for conducting thorough spatio-temporal analyses, making them particularly suited for studies aiming to unravel the complex interplay of factors influencing flight delays over time and space.

### *4.8. Discussion*

In this section, we will first explore the characteristics, applicability, and guidelines for selecting different methods. We will then provide a comprehensive analysis of the primary factors influencing flight delay propagation. Following this, we will summarize the significant challenges in current research and finally examine the impact of advanced technologies on the study of flight delay propagation modeling.

#### *4.8.1. Comparison of modeling methods*

In the previous chapters, we have conducted a detailed analysis and summarized the characteristics, applications, and suitable scenarios of various flight delay modeling methods. In reality, no single modeling approach is universally applicable to all research questions and management objectives. Therefore, in this chapter, we will perform a comparative analysis of the applicability of different modeling methods. This comparison will consider factors such as the modeling perspective, data, and research questions, aiming to establish guidance on how to select the appropriate modeling method for specific problems.

When selecting a modeling method for flight delay propagation, it is crucial to align the choice with the modeling perspective, which can be broadly categorized into flight chain-based or airport network-based analyses. Flight chain-based studies benefit from methods like agent-based simulation or economic models, which can delve into the intricacies of individual aircraft operations and financial implications. On the other hand, for a network-wide examination of airports, complex networks, and other network representation methods offer a systemic view of the interconnections and patterns across the entire network. Although queueing models are typically not applied from a flight chain-based viewpoint and economic models rarely reflect the airport network-based perspective, epidemic spreading models have the flexibility to be used in both contexts due to their ability to model the contagion-like spread of delays. Therefore, researchers should consider the modeling perspective to guide their selection of the most appropriate method, ensuring it complements the specific scale and focus of their study.

In the selection of modeling methods for flight delay propagation, aligning with available data types is paramount. Machine learning and deep learning excel with large, complex datasets, adept at revealing hidden patterns through their capacity to analyze vast and intricate data structures. Conversely, statistical models are preferred for structured, historical data, offering precise analyses of trends and patterns over time. Network representation methods, valuable for their macro-level overview of air traffic networks, require comprehensive data to effectively model network dynamics but often overlook detailed operational data like weather

**Table 14**
Summary of the characteristics and description of factors influencing flight delays.

| Factors | Variable category | Model category | Description |
|---|---|---|---|
| Airport characteristics | Dummy variable and Discrete variable | Airport network-based and Flight chain-based | Describe the number of departures, the number of landings, the size of the airport, and whether the airport is congested. |
| Delay characteristics | Dummy variable and Discrete variable | Airport network-based and Flight chain-based | Describe cumulative delays for upstream flights, average departure and arrival delays at airports, and whether delays will result in flight cancellations. |
| Weather condition | Dummy variable | Airport network-based and Flight chain-based | Describe the extreme weather at the departure airport or destination airport, such as heavy rain or fog. |
| Aircraft type | Dummy variable | Flight chain-based | Describe whether the aircraft is heavy or wide. |
| Airline information | Dummy variable | Airport network-based and Flight chain-based | Describe whether the aircraft is owned by an airline, as different airlines have different operating policies. |
| Hub status | Dummy variable | Flight chain-based | Describe whether the airport is a transportation hub. |
| Time information | Dummy variable | Airport network-based and Flight chain-based | Describe the month of the time, whether it is a weekend, and what time of day it is. |
| Flight information | Dummy variable | Flight chain-based | Describe whether it is an international flight or an overnight flight. |
| Buffer time | Discrete variable | Flight chain-based | Airlines have different flight buffers and turnaround buffers for each flight. |

This table's second and third columns further elucidate the characteristics of each factor. The first column, named "Variable category", includes three possible values: "Dummy variable", "Discrete variable", and "Dummy variable and discrete variable", to describe the type of variable of the factor. The second column, named "Model category", also contains three possible values: "Flight chain-based", "Airport network-based", and "Airport network-based and flight chain-based", to indicate the model category applicable to the factor. However, the studies we have reviewed so far only conclude with the categories "Airport network-based and flight chain-based" and "Flight chain-based".
It should be noted that the statistical analysis presented here aims to categorize the influential factors, not the articles themselves. Given that a single factor may be featured in multiple articles, a factor can fall into the category of "Airport network-based and flight chain-based". This categorization reflects the recurrent appearance of factors across various studies.

conditions or flight trajectories. This gap stems from a focus on broader network structures rather than the operational specifics, compounded by the challenges in gathering fine-grained data. Hence, researchers must judiciously select a modeling approach that not only leverages the available data but also accurately captures the dynamics of flight delay propagation. This entails a careful assessment of each method's strengths in relation to the data at hand, ensuring the chosen approach can adequately model the complexities of delay causes and propagation within the constraints of the data's scope and depth.

When selecting a modeling method for flight delay propagation, the type of research question and the temporal scale being addressed are critical factors to consider. Economic models are well-suited for analyzing financial impacts and policy implications over longer time scales. In contrast, queueing models are typically applied to address operational bottlenecks and are often used for short-term, day-to-day operational planning due to their ability to simulate daily constraints and immediate effects. Bayesian networks are adept at revealing causal relationships, making them valuable for both short-term and long-term analyses, whereas graph neural networks excel in identifying and learning complex network dynamics, which are particularly useful for medium to long-term strategic planning. Complex networks are generally used for medium to long-term analyses to assess network performance and delay patterns. On the other hand, epidemic models and queueing models are usually designed for short-term modeling, constrained by daily simulations, and tend to focus on the impact of short-term events on delay propagation. Researchers should choose a method that aligns with the specific demands of their study, considering the temporal scope and the nature of the research questions to ensure accurate and relevant results.

#### *4.8.2. Influential factors analysis*

We have summarized the following nine categories of factors according to all the articles analyzing delay propagation from the perspective of influencing factors. The relevant characteristics and descriptions are shown in Table 14. The factors influencing flight delay propagation as discussed in the literature can be broadly categorized into three types: dummy variables, discrete values, and a combination of both. The first category, using dummy variables, covers a wide range of factors. For example, extreme weather conditions like heavy snow and fog significantly exacerbate delay propagation and are often represented by single dummy variables (Brueckner et al., 2022). Similarly, the impact of periods on delays is highlighted through dummy variables, indicating how different times of the year or day affect delay propagation (Brueckner et al., 2021a). The size and type of airlines, particularly smaller and low-cost carriers, are also represented using dummy variables, as these airlines are noted to be more susceptible to delays (Brueckner et al., 2021a). Other aspects like aircraft types (Wong and Tsai, 2012), flight information (Kim and Bae, 2021), and hub status (Brueckner et al., 2022) are similarly analyzed. The second category comprises concrete numerical values, such

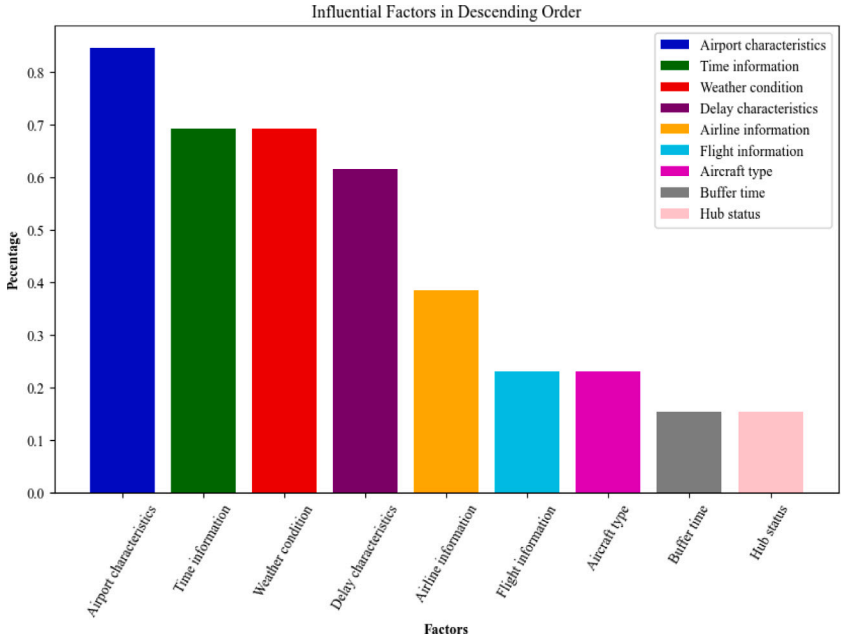



**Fig. 12.** Influential factors in descending order.

as flight and turnaround buffers (Brueckner et al., 2022). The size of these buffers varies and larger buffers tend to reduce the likelihood of delay propagation. The third category includes factors that encompass both data types. Airport characteristics, for instance, involve numerical data like the number of take-offs and landings, as well as dummy variables for specific times, indicating crowdedness (Hao et al., 2014). Delay features combine discrete variables like accumulated upstream flight delays with dummy variables for outcomes like flight cancellations (Wang et al., 2003).

We then aim to demonstrate the utility of various factors in elucidating the delay propagation across different studies. To enhance the characterization of these factors, we deliberately distinguished between modeling perspectives based on airport networks and those based on flight chains. For the weather condition factor, the former broadly examines the influence of weather conditions across a wide area (Hao et al., 2014). In contrast, the flight chain-based model offers a more granular approach, focusing specifically on the weather conditions at the departure and destination points of individual flights (Brueckner et al., 2021a). When it comes to modeling delay characteristics, airport network-based models are notably comprehensive, capturing a wide array of factors contributing to delays while flight chain-based models typically focus more on the accumulated upstream delays (Tan et al., 2021). The characteristics of airports play a vital role in both models (Lykou et al., 2020). It is a given that, regardless of the chosen model, the intrinsic attributes of an airport are factored in when it is treated as a node. This inclusion is a logical step, acknowledging the unique influence of each airport in the network. Time elements such as seasons and months are typically integrated into the airport network-based model (Nayak and Zhang, 2011). On the other hand, the flight chain-based model tends to use shorter time windows, often spanning just a few hours, to track temporal shifts (Zheng et al., 2020). Both models acknowledge the varying impacts of different times of the day – morning, afternoon, and evening – on flight delays. However, when it comes to airline-specific variables, the incorporation becomes challenging. As a result, such variables are generally exclusive to the flight chain-based model. An exception to this is (Kim and Bae, 2021), which uniquely compares airline data within the airport network framework. Lastly, factors like aircraft type (Wang et al., 2003), hub status (Brueckner et al., 2022), flight information (Kim and Bae, 2021), and buffers (Brueckner et al., 2021a) are route-specific and challenging to incorporate at the broader airport network level. Therefore, these elements are typically considered only within the flight chain-based model.

As shown in Fig. 12, we present the nine most significant factors impacting flight delays in our paper, ranked in descending order based on their prevalence. The trio of factors that most frequently emerged in our analysis are airport characteristics, time information, and weather conditions. These three factors are referenced in nearly 70 percent of the papers reviewed since they are fundamental aspects that directly affect flight operations and are universally considered in both airport network-based and flight chain-based models. The incidence of delay characteristics is also notable, being mentioned in over half of the articles. On the lower end of the spectrum, the factors of flight information, aircraft type, buffer time, and hub status are observed less frequently. This reduced presence is attributed to their specific application within the flight chain-based model, which accounts for their lower occurrence rates as compared to the other factors.

#### *4.8.3. Modeling challenges analysis*

Currently, there are numerous challenges in the field of flight delay propagation modeling. From both representative and cutting-edge literature, we have identified the following six key challenges in the modeling process.

***Challenge in Applicability and Extensibility of Delay Propagation Modeling Methods***

A significant challenge in flight delay propagation modeling is ensuring the methods' applicability and extensibility to different airports, countries, and time periods (Rodríguez-Sanz et al., 2019). Many existing models are developed for specific scenarios or datasets, such as certain US commercial airports during a limited time frame (Li and Jing, 2021). Ensuring that these models maintain their accuracy and effectiveness when applied to other contexts is a key difficulty. This challenge involves improving the model's accuracy through more comprehensive testing and methodological enhancements, as well as verifying its generalizability by applying it to various airports and comparing delay mechanisms across different countries (Du et al., 2018). Achieving this level of extensibility would provide valuable insights for alleviating delays and enhancing the effectiveness of delay management strategies in diverse air traffic systems globally (Guo et al., 2022).

***Accounting for Inter-Aircraft Delay Propagation***

The current study primarily addresses delay propagation within individual flights with the same tail number, but it does not explicitly consider delay propagation between different aircraft, which can occur due to passenger and crew connections, as well as shared airport resources. Incorporating passenger and crew data is crucial to account for the complexities of delay propagation in scenarios involving flight transfers (Kafle and Zou, 2016; Wu and Law, 2019). This process can reveal a more intricate web of delay propagation. In this context, one of the significant challenges is distinguishing between delays that are propagated from other flights and those that are newly generated by the aircraft itself (Kafle and Zou, 2016). This distinction is difficult due to limitations in existing data. Future research should explore methods to integrate these additional data types, such as passenger and crew information, to enhance the accuracy of models in capturing the full dynamics of delay propagation, particularly in the context of connecting flights (Guleria et al., 2019).

***Challenge in Modeling Comprehensive Operational Factors for Accurate Delay Propagation***

A significant challenge in flight delay propagation modeling lies in incorporating a full spectrum of operational factors affecting both the ground and airborne flight operations (Lin et al., 2021). This includes capturing the complexities of airport runway operations, variations in runway configurations, and the diverse activities of airspace users (Zhao et al., 2022). Accurately integrating these factors is crucial to model the intricacies of real-world flight dynamics comprehensively. Overcoming this challenge involves developing models that can simulate the multifaceted interactions and impacts of these operational elements, thereby enhancing the accuracy of delay predictions and the effectiveness of delay management strategies within the air transportation system (Bao et al., 2021; Cai et al., 2021a).

***Challenge in Analyzing Time-Varying Network Topologies in Flight Delay Propagation***

A primary challenge in modeling flight delay propagation using complex network analysis lies in understanding and evaluating the robustness and resilience of air traffic networks with time-varying topologies (Gopalakrishnan and Balakrishnan, 2021). Traditional network models often assume static network structures, but in reality, the topology of air traffic networks is dynamic, changing over time due to various operational, environmental, and regulatory factors (Li and Jing, 2021). This dynamic nature complicates the analysis, as it requires models to account for the temporal variations in network connections and interactions. Future research needs to develop methodologies and tools that can effectively capture and analyze these time-varying characteristics of air traffic networks, thereby improving the understanding of how delays propagate through an ever-evolving network structure (Du et al., 2018; Wu et al., 2022).

***Challenge of Model Adaptation to Data Drift***

Flight delay models are typically built on historical data, often with the assumption that data distributions remain constant over time. However, the operational context of airlines and airports can change due to a variety of factors, including shifts in the real-world environment, laws, regulations, economic conditions, and health crises. This issue, known as "data drift", requires models to continually adapt while retaining their effectiveness (Carvalho et al., 2021). The aviation industry is constantly evolving, influenced by advancements in science and technology. This evolution presents a substantial challenge for researchers in developing models that are not only accurate but also flexible and responsive to these dynamic changes.

***Enhanced Data Collection for Detailed Analysis of Delays***

One of the primary challenges in flight delay propagation modeling is the need for enhanced data collection that encompasses a wider range of data types, including cargo flights, air traffic control information, and other unpredictable events such as unscheduled maintenance (Kafle and Zou, 2016; Yu et al., 2019; Sun et al., 2022). These studies underscore the importance of gathering more detailed and comprehensive data to facilitate a more accurate analysis of factors contributing to delays, such as ground turnaround times and their influence on delay propagation. By improving data granularity, including capturing details of events that are often difficult to predict, researchers can better differentiate between various types of delays and reduce potential errors in econometric models (Kafle and Zou, 2016). This improved data collection is crucial for developing a nuanced understanding of the causes of delays and the intricate process of their propagation within the air transportation system (Pyrgiotis et al., 2013; Cai et al., 2021a).

#### *4.8.4. Analysis of advanced technologies*

***Application of machine learning and deep learning technologies***

In the context of air traffic delay networks (Ai et al., 2019) or airport situation maps (Shao et al., 2022), the data can be gridded to facilitate spatial feature extraction using CNNs. This process involves transforming delay data or aircraft trajectory information into a grid format, which CNNs can then analyze to learn spatial characteristics of the delay distribution. Recurrent neural networks (RNNs), especially LSTM models, are well-suited for processing time series data, such as the sequence of delay states at airports (Zhang et al., 2021a; Zeng et al., 2021; Bao et al., 2021). By capturing the temporal dependencies in delay sequences, these networks can effectively learn the dynamics of delay propagation over time.

With the rise of graph deep learning, various GNN models have been widely applied in spatio-temporal traffic prediction tasks (Bui et al., 2022). In the construction of airport networks, where nodes represent airports and edges represent flight connections, GNNs can effectively capture the complex, non-Euclidean relationships within the network (Cai et al., 2021a; Wu et al., 2022). Besides airport networks, flight chains (Guo et al., 2020) and O–D pairs (Rebollo and Balakrishnan, 2014) can also be conceptualized as nodes in a network. Different metrics can be used to measure node similarity, revealing various network characteristics (Jiang and Luo, 2022). Similar graph neural network techniques can be applied to these network structures to predict delays. This approach allows for a more nuanced understanding of delay propagation, considering not just the spatial layout of airports but also the relational dynamics of flights and passenger flows.

***Integration with existing methods***

Indeed, many studies have adopted a combination of diverse methodologies to better understand and model flight delay propagation in large-scale aviation networks. These studies often merge different approaches to achieve more comprehensive results and insights. For example, researchers may combine queueing models or epidemic spreading models with simulation-based models for validation experiments in large aviation networks (Pyrgiotis et al., 2013; Lin et al., 2021; Baspinar and Koyuncu, 2016; Ceria et al., 2021). They may also combine statistical models with economic regression techniques for significance testing to identify crucial factors influencing delay propagation (Wong and Tsai, 2012; Kafle and Zou, 2016; Tan et al., 2021). Additionally, after constructing networks using different methods like Bayesian networks or the Granger causality test, researchers may use complex network indicators for network performance analysis (Rodríguez-Sanz et al., 2018; Li and Jing, 2021; Du et al., 2018).

Machine learning models can also be integrated into well-established statistical or economic models to enhance both performance and interpretability. For instance, machine learning techniques can be employed to predict delays in preceding flights and then input these predictions into models that analyze the connections between successive flights (Chen and Li, 2019). This iterative approach enables more accurate forecasting of delays for chained flights, leveraging the predictive capabilities of machine learning while maintaining the structural insights provided by economic models. In agent-based modeling, machine learning techniques can be utilized to estimate flight times based on historical data (Cai et al., 2021b). This approach allows for more accurate simulations of flight operations, as machine learning algorithms can learn from past data to predict future outcomes, thereby enhancing the realism and reliability of agent-based models. Queueing models, often used in operational research, can be augmented with machine learning algorithms to estimate operational parameters more efficiently. This integration can significantly speed up simulations by providing rapid and accurate estimations of key parameters, such as service times and arrival rates, which are critical for modeling airport operations and managing delays.

It is crucial to acknowledge that while many studies employ a blend of methodologies, our analysis focuses on the core methodologies used, discussing each primarily within a single approach for clarity. This approach recognizes the varied analytical perspectives and scopes of application inherent in this field, highlighting that there is no universally applicable modeling method. Researchers are encouraged to select methodologies that align with their specific research questions and objectives. The key takeaway is the value of methodological diversity. Employing a combination of approaches can lead to a more comprehensive understanding of flight delay patterns, thereby informing strategies to effectively mitigate delays and enhance the overall efficiency of the aviation industry.

***Limitations and frontier topics of advanced technologies***

Deep learning models, while powerful, often require extensive datasets and significant computational resources, which can be a limiting factor in their application. Additionally, the interpretability of these models is a notable challenge. While deep learning can offer superior predictive performance, its "black-box" nature can hinder its acceptance in practical scenarios where understanding the decision-making process is crucial. One effective approach to address the interpretability issue is to integrate deep learning models with traditional models known for their strong explanatory power. Techniques like SHAP (Shapley Additive exPlanations) (Lundberg and Lee, 2017) analysis are widely used for post-hoc interpretation, providing insights into the importance of features used by the model.

Integrating diverse types of data, such as weather information, flight schedules, and real-time operational data, can significantly enhance the accuracy of delay models. Moreover, mining causal relationships from delay data can provide deeper insights into the dynamics of flight delay propagation, aiding in the development of more informed policies and strategies. In the subsequent Section 5, we will delve into these limitations and frontier topics in greater detail, discussing how they shape current research and future directions in the field of flight delay propagation modeling.

## 5. Current trends and future opportunities

From our literature review, it is evident that numerous research methods have been employed to explore flight delay propagation patterns from various modeling perspectives. However, there remain several challenges and unresolved issues. In this section, we analyze current research trends from a statistical viewpoint and discuss future research opportunities.

### *5.1. Statistical analysis and current trends*

Fig. 13 illustrates the number of studies on various flight delay propagation methods from 1999 to 2022. The early research period features fewer studies and more dispersed methods, with no single dominant research approach. However, in recent years, there has been a significant upward trend in the number of studies, particularly as the total number of articles in the last five years surpasses the combined sum of articles from 1999 to 2017. Notably, there is a sharp increase in network representation

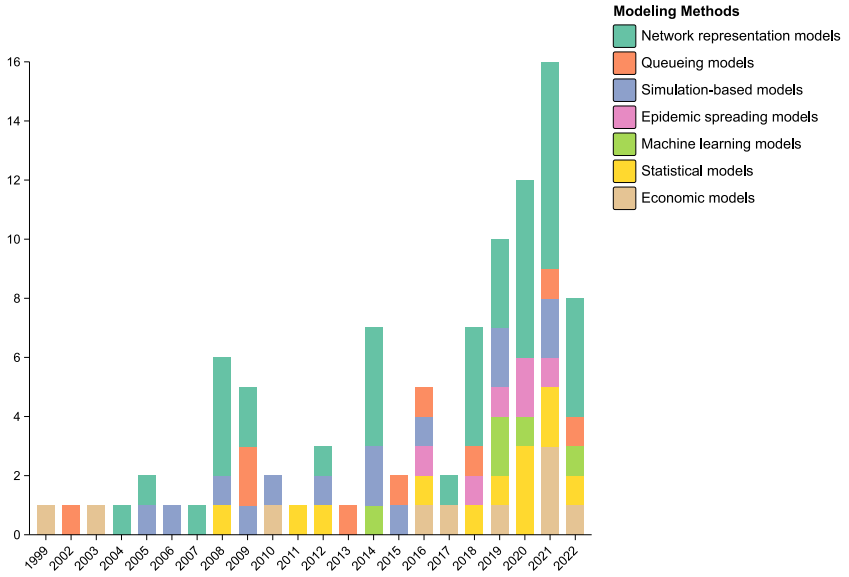



**Fig. 13.** Number of publications by year with respect to different modeling methods.

methods in 2008 and 2009, with many conference papers during these years exploring flight delay propagation from a Bayesian network perspective. As the network effect of delay propagation becomes increasingly evident and garners attention, the number of articles utilizing network representation methods as modeling tools has grown at a faster pace. It is important to mention that while composing this literature review, articles published after September 2022 are not included in our statistical analysis.

Fig. 14 provides a more specific illustration of research trends in network representation by showing the proportions of various network representation methods. Bayesian networks and complex networks are the most commonly employed methods for analyzing delay propagation patterns within network scopes. By constructing network structures of interest, Bayesian networks can probabilistically represent the impact of different factors on delay propagation, while complex networks can offer a range of performance indicators for network analysis. Furthermore, since alternative spatio-temporal methods encompass a variety of research approaches, they also occupy a larger proportion. It is worth noting that, given the strong network representation capabilities of graph neural networks (GNNs) in recent years, networked delay modeling methods based on GNNs are likely to become an essential research tool in the future.

Fig. 15 displays the matching relationships between various data categories, modeling perspectives, and research methods. It is quite evident that flight operation data is the most widely used data type and is employed in all studies, followed by weather data and flight trajectory data. In contrast to Table 2, we also list two types of non-public data here, such as air passenger data and airport operation data. We remark that, the total numbers in the middle and right columns are not equal, as some studies use multiple data types, resulting in inconsistencies in the sum of studies based on the flight chain and airport network. In reality, the number of studies in these two modeling perspectives is not significantly different. Furthermore, there exist some mapping disciplines between modeling perspectives and research methods. For instance, queueing models are rarely analyzed from a flight chain perspective, and airport network-based modeling is seldom reflected in economic analysis methods.

### *5.2. Future opportunities for modeling methods*

#### *5.2.1. Spatio-temporal models*

Spatio-temporal graph neural networks are specifically designed to capture both spatial and temporal dependencies in graph-structured data, making them an ideal choice for modeling complex transportation networks (Jiang and Luo, 2022). By incorporating spatio-temporal information, the ST-GNN model can more effectively learn and represent the relationships between different nodes and their attributes in transportation systems, leading to enhanced performance in various traffic forecasting tasks, including flight delay prediction and propagation modeling. Despite the progress made in modeling airport network delay propagation from spatial and temporal perspectives, several advanced modeling techniques remain to be explored for a deeper understanding of the propagation principles:

(i) Coupled spatial and temporal dependencies: Delay propagation in airport networks is influenced and amplified by flights connecting different airports. Consequently, the delay propagation between two airports is contingent on the flight's travel time and the distance between airports, intertwining spatial and temporal dependencies.
(ii) Heterogeneous airport networks: Accounting for the heterogeneity of airport nodes, stemming from differences in operational levels, such as infrastructure and capacities, can lead to more accurate modeling of the spatial and temporal framework.

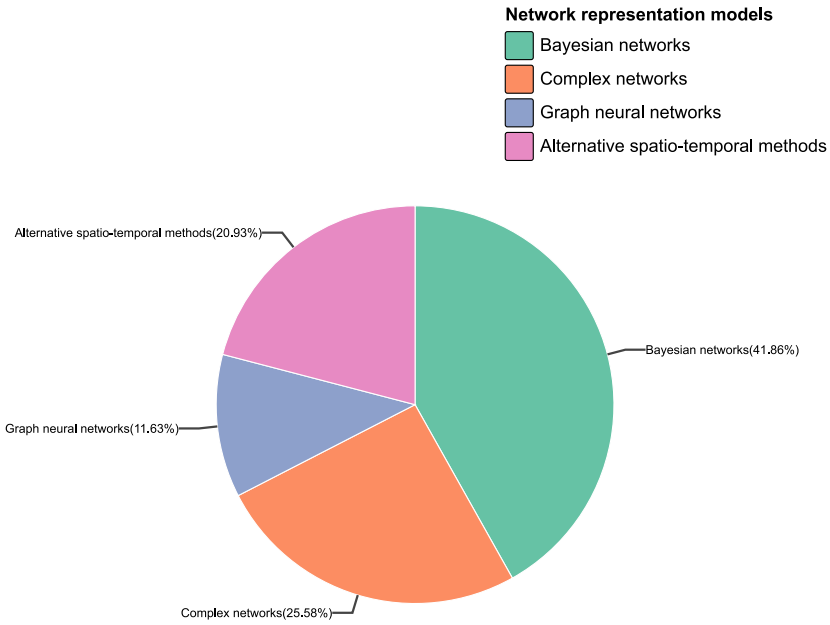



**Fig. 14.** Proportions of different types of network representation models.

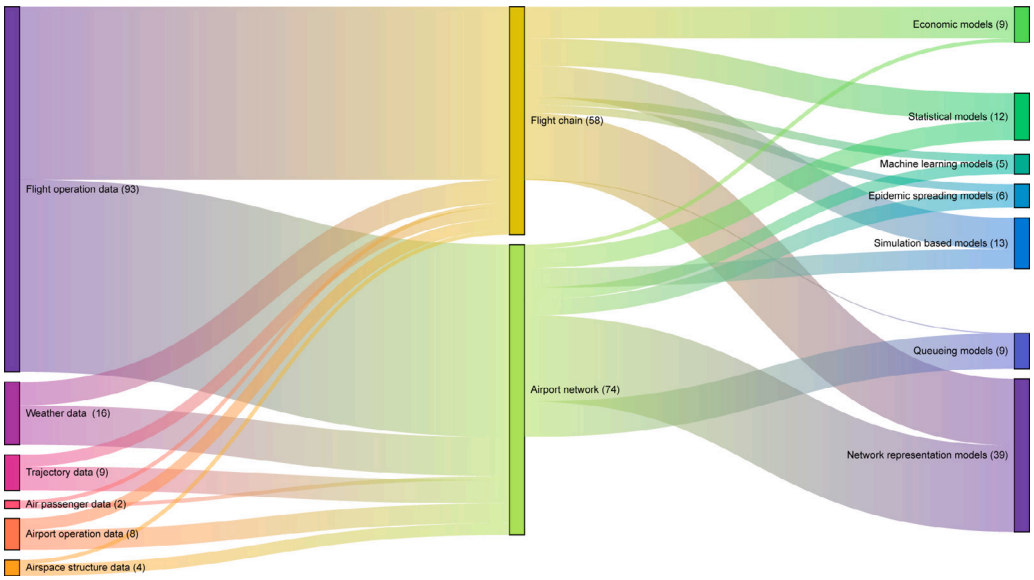



**Fig. 15.** Mapping of data category, modeling perspectives to modeling methods.

(iii) Dynamic graph structures: While existing studies largely consider static network structures, aviation networks are time-varying due to evolving flight plans, necessitating the incorporation of dynamic graph structures.

(iv) Multiple relational graph structures: Aviation networks often exhibit multiple relationships between airport nodes, such as distance relations, flight schedules, and causal relations. Developing models that consider multi-relational graph structures can provide a more comprehensive understanding of the complex interactions in delay propagation.

### *5.2.2. Causal relations*

Spatial–temporal flight delay propagation modeling constitutes a prevalent approach in gauging the correlation of delay time series from a network science perspective. Beyond spatial and temporal correlations, causal correlations are also widely employed in delay series between airport pairs, examining whether a group of time sequences engenders another set of time sequences, thus revealing inherent associations (Du et al., 2018). A plethora of techniques exist for estimating causal impacts, including spatial

econometrics models (Cheng et al., 2019), time delay stability method (Wang et al., 2019), and the frequently utilized Granger causality test (Zanin et al., 2017; Du et al., 2018; Mazzarisi et al., 2020). More recently, Guo et al. (2022) posit an advanced delay causality predicated on the non-strong and independent assumption.

Nevertheless, the Granger causality test assumes linear time serial dynamics and employs a vector autoregressive regression (VAR)-based methodology, rendering it incapable of addressing nonlinear dependencies. Existing inspection methods for studying non-linear dependencies generally utilize supplementary models, which may overlook crucial non-linear interactions between predictive factors. Consequently, merging neural networks (Tank et al., 2018, 2021) with the classical Granger causality test presents a promising research trajectory, capable of discerning both nonlinear interactions and extracting the Granger causal structure. To elucidate the causality dynamics within the air transportation system, contemporary research endeavors have employed alternative causal inference methodologies. By constructing structural causal networks from empirical data, these studies scrutinize the intricate causal interrelationships among airport variables and delay occurrences within the complexities of the system (Truong, 2021; Zeng et al., 2022).

#### 5.2.3. Explainable models

In recent years, the ubiquity of extensive data, computational efficacy, and scientific advancements have spurred a proliferation of research efforts developing machine learning and deep learning methodologies for flight delay and airport network delay propagation prediction. However, the challenge of model interpretability remains a salient critique for "black-box" machine learning or deep learning models, encompassing delay propagation prediction and other traffic forecasting tasks. In the realm of aviation, it is crucial to recognize that model interpretability frequently takes precedence over performance (predictive accuracy). The rationale behind this emphasis lies in the fact that a highly interpretable model can better facilitate decision-making through its internal mechanisms. As a result, aviation stakeholders can devise more efficacious delay mitigation strategies by comprehending the underlying processes of the model.

While significant strides have been made in visualizing and elucidating other deep neural network architectures, such as CNNs, the evolution of post-processing techniques for interpreting GNN predictions remains in its nascent stages (Ying et al., 2019). Furthermore, the application of these methodologies to the traffic forecasting domain has yet to be explored (Jiang and Luo, 2022). The dearth of model interpretability may pose a more acute issue in the aviation domain compared to analogous forecasting problems in other fields. Potential avenues for exploration include the integration of GNN-based approaches with "white-box" techniques, encompassing statistical models, economic analytical models, and queueing models. Alternatively, the fusion of prior knowledge may bolster explainability in existing work (Beckh et al., 2021). One can incorporate knowledge into the machine learning pipeline, embed knowledge within the explainability method, or derive knowledge from explanations.

### 5.3. Future opportunities for research questions

#### 5.3.1. Pandemic impacts

The aviation sector is notably susceptible to the adverse effects of pandemics, as demonstrated by past events such as the SARS outbreak and the continuing COVID-19 pandemic (Colizza et al., 2006; Brockmann and Helbing, 2013; Sun et al., 2021). We elucidate the detailed influence of pandemics on the air transportation industry by the example of ongoing COVID-19. The current outbreak is an unparalleled event in air transportation, also regarded as a unique type of supply chain disruption risk (Ivanov, 2020). As of July 2022, over 577.8 million confirmed cases and more than 6.42 million fatalities have been attributed to COVID-19. The pandemic is estimated to have precipitated the largest global recession since the Great Depression in the 1930s, with the aviation sector being among the hardest hit (Suau-Sanchez et al., 2020).

Besides, numerous researchers have devoted their efforts to understanding the impact of COVID-19 on air transportation network structure. Sun et al. (2020) delve into the spatial–temporal evolutionary dynamics of COVID-19 in air transportation networks, providing a comprehensive empirical analysis of the pandemic's influence on aviation using network science tools. Li et al. (2021c) explore the ramifications of country-to-country disconnection due to varied policy interventions on air transport network connectivity during COVID-19 periods. Additionally, Li et al. (2021b) uncover the spatio-temporal variation of the worldwide air transportation network induced by the pandemic.

It is evident that the air transportation network has undergone substantial changes as a result of undesirable flight suspensions, subsequently leading to variations in flight delay propagation mechanisms. Mitsokapas et al. (2021) present a statistical analysis of arrival delays, highlighting a dramatic shift during the COVID-19 pandemic. These studies demonstrate that the most immediate consequences of a pandemic involve the reduction of flights and alterations to the aviation network structure. Simultaneously, the unpredictable nature of travel restrictions issuance further complicates the assessment of delay propagation during such periods. Thus, considering the scarcity of existing work, integrating pandemic impacts into flight delay propagation modeling is highly anticipated and appreciated.

#### 5.3.2. Passenger-centric delay

Passenger-centric delay encompasses the notion of assessing flight delays from the passengers' viewpoint, rather than concentrating solely on individual flights. This approach often arises from flight cancellations and missed connections (Barnhart et al., 2014). Passenger-centric delay models offer a more accurate representation of the overall impact of flight delays on passengers, proving valuable for airlines, airports, and regulators in understanding the true cost of delays and devising strategies to enhance the overall passenger experience (Monmousseau et al., 2019). Distinct differences exist between flight-centric and passenger-centric

delays, such as modeling focus and objectives. Additionally, from an optimization perspective, optimal flight operations do not align with optimal passenger operations outcomes when itineraries connect multiple flights, resulting in a trade-off between flight and passenger delay operations (Jacquillat, 2022).

The generation and propagation of passenger-centric delay within the network are closely related to the challenges of managing passenger itineraries and corresponding connections (Cook et al., 2015). However, passenger itinerary data is generally inaccessible to the public. Thus, various methods have been developed to estimate passenger itineraries. Barnhart et al. (2014) utilize the multinomial logit model for estimation while Jacquillat (2022) propose a statistical learning model to predict itineraries in aviation networks, achieving satisfactory performance. Alternatively, Monmousseau et al. (2019) leverage publicly available data, i.e. Twitter data, containing valuable information for estimating the passenger-centric delay.

Although passenger-centric delay and flight-centric delay address distinct aspects of delay propagation in the air transportation system, incorporating elements of passenger-centric delay modeling can augment the understanding of flight-centric delay propagation and overall system performance. Monmousseau et al. (2019) demonstrate that employing a passenger-centric feature set yields superior predictive accuracy for flight-centric air traffic perturbation values on a system-wide level compared to using forecasting tools on flight-centric data alone. Besides, incorporating passenger itineraries, connection times, and rerouting information into the construction of a passenger network, in conjunction with the flight delay propagation network, can lead to a more comprehensive representation of delay propagation mechanisms.

#### *5.3.3. Multi-layer network delay*

Multi-layer air transportation networks, comprising multiple interconnected layers, offer a sophisticated and comprehensive approach to modeling and analyzing the complex dynamics of air transportation systems. Due to the hierarchical structure of aviation networks with diverse attributes, such as flight connections, passenger itineraries, and airport infrastructure, representing specific system components, there are multiple perspectives from which to examine the interrelationships between different sublayers and identify the dynamic evolutionary characteristics.

Du et al. (2016) investigate the geographical and demographic features of the Chinese Airline Network (CAN) and use the K-core decomposition method to portray the network in a stratified fashion. The analysis uncovers similarities between cities and regions within the same layer, substantiating the existence of a hierarchical structure in the CAN. Jiang et al. (2016) categorize air transportation networks into collections comprising various airlines, with each airline representing a layer. By employing network aggregation methods and integrating structural indicators, they examine the evolutionary characteristics of the CAN. This layer-decomposition technique is expanded upon by Guo et al. (2019), who propose a multi-scale network science approach for predicting flight volume and identifying community structures. In recent research, Rodriguez-Deniz et al. (2022) use a probabilistic latent model to explore the dynamic evolutionary features of multilayer networks. Experimental results indicate that the model possesses strong predictive capabilities for network connectivity and community detection.

Delay propagation across different layers mirrors the interdependencies and interaction traits between various aviation network layers. Giannikas et al. (2022) construct a multi-layer network composed of aircraft, crew, and passengers, analyzing how flight delays propagate throughout the network. They demonstrate the value of incorporating additional layers of information in delay propagation analysis, providing comprehensive network visibility over delay propagation. Consequently, studying delay propagation in these constructed multi-layer air transportation networks can more effectively capture the propagation characteristics of delays at different layers. This allows for a finer-grained examination of delay mechanisms, aiding aviation decision-makers in devising more targeted delay management measures.

#### *5.3.4. Data heterogeneity*

Data heterogeneity refers to the inconsistencies and differences that can be observed in datasets due to various factors, such as differing formats, data sources, or data collection methods (Jiang and Luo, 2022). In the context of flight delay propagation modeling, these inconsistencies can arise from multiple factors:

(i) Diverse data sources: Flight delay data typically originates from various sources, such as airlines, airports, air traffic control departments, aviation management departments, and third-party data organizations. These data are collected and reported in a variety of ways, resulting in data heterogeneity.
(ii) Multi-modal data: Flight delay data also exhibits multi-modal characteristics. For instance, there may exist image data (radar cloud maps), text data (weather reports), and numerical data (flight plan data, trajectory data, airport and airspace structure data) during the data processing.
(iii) Measurement discrepancies: Different entities might employ distinct methods or definitions for measuring flight delays, leading to inconsistencies in the data.
(iv) Data quality: Data quality concerns include missing, noisy, and sparse data in delay propagation modeling. In some cases, flight delay data might be incomplete or missing due to various reasons, such as data collection errors and communication issues.
(v) Data privacy: As highlighted in Chung et al. (2020), flight delay data is highly sensitive, particularly when it involves airline operational performance and passenger personal information.

Therefore, to address the aforementioned data heterogeneity issues, it is crucial to preprocess the flight delay data initially. For instance, for multi-modal data, data fusion and data integration techniques can be employed before proceeding with subsequent forecasting tasks to enhance the robustness of the predictions (Holzinger et al., 2021). Additionally, to tackle data privacy concerns, data encryption methods or differential privacy techniques can also be implemented (Chung et al., 2020). Indeed, through rigorous data processing, we can gain a more accurate and comprehensive understanding of flight delays. By addressing data heterogeneity issues and ensuring data quality, researchers and aviation decision-makers can develop more precise models and implement targeted strategies for managing delays and improving the overall efficiency of air transportation systems.

#### *5.3.5. Buffer time and delay mitigation*

Buffer time generally consists of flight buffer and ground buffer, which pertain to the supplementary duration allocated to the shortest feasible flight time or aircraft turnaround time in order to minimize late flight arrivals and increase the likelihood of on-time departures as per the scheduled takeoff time (Brueckner et al., 2021a). Ground buffer, which deals with connections between consecutive flights, can effectively reduce delay propagation when set reasonably (Kafle and Zou, 2016). Moreover, the optimization and mitigation of propagated delays can be achieved by considering other airline scheduling challenges, such as aircraft routing and crew pairing, in light of delay propagation (Dunbar et al., 2012; Liu et al., 2016). As a result, efficient delay propagation modeling and estimation can illuminate the underlying mechanisms of delay propagation and inform the development of optimal delay mitigation strategies. Conversely, buffer time and delay mitigation strategies address uncertainties and complexities inherent in the air transportation system. A more comprehensive understanding of these factors is expected to contribute to the improvement of flight delay propagation modeling (Wang et al., 2022b).

In relation to buffer time, Brueckner et al. (2021a, 2022) explore the impact of schedule buffer time on delay propagation levels through extensive theoretical and empirical analyses. Additionally, Wang et al. (2022b) determine statistically significant time lags for airport delay propagation using noisy and aggregate operational data. These valuable insights into buffer time and delay propagation timescales enable a more accurate representation of flight operations, enhancing system predictability and reliability, which may subsequently inform flight delay propagation modeling.

To tackle the cascading nature of flight delay networks, Niu et al. (2021) employ the CARP method for dynamics modeling, which is grounded on Markovian alternating renewal processes. Concurrently, Gopalakrishnan and Balakrishnan (2021) utilize the Markov jump linear system model to capture the delay propagation characteristics and network properties. Notably, both studies incorporate various control measures into their respective flight delay propagation models and assess the impact of these measures on network delay propagation. Consequently, the integration of delay mitigation strategies into delay propagation models effectively captures the uncertainty and complex characteristics of the air traffic system during the delay propagation process, while also facilitating the development of proactive strategies to reduce the overall delay level within the aviation network.

## 6. Conclusion

In this paper, we provide a comprehensive review of flight delay propagation modeling. Our review begins with an enumeration of datasets employed in the studies, providing detailed statistics on data sources, data categories, availability, and usage frequency in various air transportation systems. Concurrently, we also present the application scenarios of different data types in delay propagation modeling. Subsequently, we introduce the two most common modeling perspectives for delay propagation, namely flight chain-based modeling and airport network-based modeling. We then delve into a thorough analysis of seven distinct research methods used in delay propagation studies, highlighting their characteristics and application scopes. This includes a comparison of how specific methods are utilized across different studies, along with a discussion on the application of frontier technologies in this domain. Finally, we summarize the current research trends from a statistical analysis perspective and outline future research opportunities from the standpoint of modeling methods and research questions. We summarize our findings on flight delay modeling with respect to the three proposed research questions.

**Data types and application scopes:** In flight delay propagation modeling, the most extensively used and essential data is flight operation data, followed by weather data and flight trajectory data. Flight operation data encompasses estimated and actual departure and arrival times necessary for calculating delays, providing the foundation for understanding delay patterns. Meanwhile, weather data and flight trajectory data can act as supplementary features or information input to the model to enhance its predictive capabilities. Regarding flight operation data, the BTS dataset is the most widely used, primarily due to its comprehensive public accessibility. Additionally, many studies acquire data from specific airports, airlines, and aviation management departments, which is typically challenging to access publicly. However, this specialized data commonly offers more in-depth insights into the operations of individual airports, airlines, or specific regions, which can be beneficial for targeted analysis. For weather and trajectory data, the NOAA and Flightradar24 data are popular choices, as they are also open to the public. In our comprehensive analysis, we have identified distinct patterns in the use of data sources across various air transportation systems, ranging from local to regional scales. This investigation reveals how the scale of operations, in terms of the number of flights and airports involved, influences the choice and application of data sources. Our findings provide valuable insights into the diverse methodologies and approaches adopted in different systems, reflecting the unique challenges and requirements at each level of air transportation. Different data types are indeed applied to various levels of delay propagation modeling, and incorporating additional data in specific scenarios is crucial for a comprehensive understanding of delay propagation patterns. For instance, airspace structure data, which includes information on airspace boundaries, air routes, and air traffic control sectors, is indispensable in multi-airport systems and micro-level regional delay propagation modeling.

**Modeling methods and their characteristics:** We have indeed discussed the characteristics and application scopes of seven major categories of methods used in flight delay propagation modeling, including economic analytical models, statistical models, machine learning models, epidemic spreading models, simulation-based models, queueing models, and network representation models. Different types of methods generally target specific research questions, and their applicability depends on the context and objectives of the study. For example, complex networks are typically analyzed from a medium and long-term perspective, aiming to understand the structure and behavior of flight delay propagation across airport networks over extended periods. In contrast, epidemic spreading models and queueing models focus on delay propagation within a day, providing insights into short-term dynamics and the impact of congestion or specific events on flight delays. These methods also align with specific modeling perspectives. Network representation methods usually concentrate on airport network-based modeling, capturing the relationships and dependencies among airports and the overall system's dynamics. On the other hand, economic analytical models are more concerned with flight chain-based modeling, emphasizing the individual flight segments and their connections in terms of delay propagation. Ultimately, the choice of research method depends on the research question, the available data, and the desired level of analysis. Researchers should carefully consider these factors when selecting the appropriate modeling technique for their specific study of flight delay propagation. In the discussion section, we delve into the characteristics and applicability of various modeling methods. We provide guidance on selecting modeling approaches based on data considerations, modeling perspectives, and research questions. We then outline the nine major factors influencing flight delay modeling in current research and the six primary modeling challenges. Lastly, we discuss the role of advanced technologies in flight delay modeling, focusing on three main areas: the use of machine learning and deep learning technologies, their integration with traditional methods, and addressing limitations while exploring new frontiers. These discussions aim to underscore the study's significance and its potential implications for researchers and stakeholders, ensuring a comprehensive understanding of the field's current state and future directions.

**Current trends and future research opportunities:** In recent years, the number of papers studying flight delay propagation has indeed been gradually increasing. In contrast, earlier research is relatively scarce, and the methods employed are more dispersed. As the network effects of flight delay propagation become more evident, network representation methods are receiving widespread attention, offering a robust approach for capturing the complex interactions within the aviation system. Bayesian networks and complex networks are two classic and widely used network representation methods. Bayesian networks provide a probabilistic framework for understanding the relationships and dependencies among various factors affecting flight delays, while complex networks enable the analysis of the topological characteristics and performance indicators of airport networks. Additionally, with the development of cutting-edge technologies such as big data and deep learning, research on network representation methods based on graph neural networks is rapidly advancing. Graph neural networks have the potential to further improve the modeling and understanding of flight delay propagation patterns by leveraging the inherent structure and properties of network data. Moreover, many recent studies involve the combination of multiple methods to better explore the rules of flight delay propagation. By integrating different modeling techniques, researchers can develop more comprehensive and accurate models that capture the intricacies of flight delay propagation from various perspectives.

Furthermore, we have outlined relevant research opportunities based on modeling methods and research questions. These research potentials, stemming from modeling methods, can be summarized from three perspectives: spatio-temporal methods, causal relations, and explainable models. Spatio-temporal methods emphasize the importance of considering various elements, such as dynamic and multiple relational graph structures when using the ST-GNN model. Causal relations provide deeper insights into the underlying factors that contribute to flight delays while developing explainable models is crucial for providing profound guidance in mitigating flight delays, as it allows stakeholders to understand the reasons behind model predictions and make informed decisions. Additionally, we contemplate prospective research questions from diverse standpoints for future exploration. For instance, examining delay propagation within the intricate and dynamic milieu engendered by pandemics, as well as confronting challenges in data dimensions such as multi-modal data and data privacy concerns. Moreover, by scrutinizing delay propagation from the perspectives of passenger-centric delay, multi-layer network delay, and delay mitigation, a more profound comprehension of the complex interrelations, hierarchical structures, and indeterminate elements within the delay propagation process may be attained, thereby facilitating more comprehensive modeling endeavors.

Our comprehensive review provides findings of immense value to researchers and stakeholders in the field. The significance of our work, from "Impact" and "Application" perspectives, is summarized in the following five areas:

1. **Comprehensive Literature Review:**
    - *Impact:* The review streamlines the vast amount of literature on flight delay propagation modeling, making it more accessible for researchers and practitioners. This organization of knowledge allows for faster dissemination and adoption of existing findings and methodologies.
    - *Application:* A comprehensive review provides a solid foundation for researchers and practitioners to build upon, allowing them to develop novel methods or improve existing ones, and apply these methods to real-world problems in logistics, transportation, and supply chain management.
2. **Data Source and Usage Analysis:**
    - *Impact:* Understanding the data landscape in flight delay propagation modeling helps researchers and practitioners identify gaps and challenges related to data collection, processing, and utilization, thus inspiring new methodologies to address these challenges.

- *Application:* Insights into data sources and usage can enable practitioners to choose appropriate datasets for their specific applications, which can lead to better decision-making and optimization of operations in logistics, transportation, and supply chain management.

3. **Modeling Perspectives:**

   - *Impact:* Differentiating between flight chain-based and airport network-based modeling offers researchers and practitioners a clear distinction between these two perspectives, enabling them to choose the most suitable approach for their particular problems.
   - *Application:* By selecting the appropriate modeling perspective, researchers and practitioners can develop more accurate and effective models, which can then be applied to enhance operations, optimize resource allocation, and reduce delays in logistics, transportation, and supply chain management.

4. **Comparison of Research Methods**

   - *Impact:* Comparing various research methods helps researchers and practitioners understand the strengths and weaknesses of different approaches, which in turn guides them in selecting the most suitable methods for their research questions or practical applications.
   - *Application:* A thorough understanding of different modeling techniques allows researchers and practitioners to develop and apply more effective and targeted solutions to address flight delay propagation problems, leading to improved operational efficiency, reduced costs, and enhanced customer satisfaction in logistics, transportation, and supply chain management.

5. **Research Trends and Future Opportunities:**

   - *Impact:* Identifying research trends and future opportunities directs researchers and practitioners towards promising areas of investigation, fostering the development of innovative solutions and methodologies.
   - *Application:* By focusing on emerging research opportunities, researchers and practitioners can contribute to the development of new strategies, tools, and techniques that can be applied to tackle flight delay propagation challenges in logistics, transportation, and supply chain management, ultimately leading to more efficient and resilient systems.

In summary, the significance of these contributions lies in their ability to guide and inspire researchers and practitioners in developing and applying more effective, targeted, and innovative solutions to flight delay propagation problems. These solutions can have far-reaching impacts on logistics, transportation, and supply chain management, ultimately leading to improved operational efficiency, reduced costs, and enhanced customer satisfaction.

## CRediT authorship contribution statement

**Chi Li:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Jianfeng Mao:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Lingyi Li:** Writing – review & editing, Writing – original draft, Visualization, Methodology. **Jingxuan Wu:** Writing – review & editing, Validation, Methodology. **Lianmin Zhang:** Writing – review & editing, Supervision. **Jianyu Zhu:** Writing – original draft, Visualization. **Zibin Pan:** Investigation, Data curation.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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